# Precise Localisation for Vineyard Pruning Operations

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# ABSTRACT

This thesis is bloody brilliant.

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Thanks mum and dad!

# PREFACE

A preface is useful for chatting to the reader about why you wrote this thesis, what is original about your work, and what material you have published.

# NOTATION

If you have a lot of mathematics it is a good idea to have a notation section and a glossary of the symbols employed in the thesis. In this section you might also include definitions and acronyms. The LaTeX package nomencl is useful for constructing a glossary.

# Chapter 1

# INTRODUCTION

## 1.1 RESEARCH CONTEXT

This thesis is situated within the broader research initiatives of the University of Canterbury Computer Vision Research Group (UCVision), specifically contributing to 'The Occlusion Project'. This project utilises autonomous field robots to capture multi-view imagery of vineyards and orchards, which are subsequently processed *offline* to generate high-fidelity 3D reconstructions using 3D Gaussian Splatting (3DGS). These models serve as the "digital twin" required for complex agricultural tasks such as precision yield estimation and automated labour.

A primary objective of this project is the automation of vineyard pruning. In this workflow, pruning decisions and robotic arm path planning are computed within the high-fidelity 3DGS environment. Unlike conventional pruning systems that rely on lower-resolution, real-time sensor data for immediate modelling and execution, the UCVision workflow decouples data collection from intervention. While this allows for superior structural analysis of the vine, it introduces a critical spatial synchronisation problem: the realignment of global and local coordinate frames.

For the robot to execute a pre-computed pruning plan, it must bridge the gap between its real-time physical state and the coordinate system of the offline model. This necessitates a robust SLAM framework capable of solving three distinct sub-problems:

1. **Global Relocalization:** Navigating the platform back to the specific plant or row identified in the modeling phase.
2. **Strategic Positioning:** Aligning the mobile base within the physical workspace constraints of the robotic arm to ensure all required cutpoints are reachable.
3. **Sim-to-Real Registration:** Achieving millimeter-level alignment between the robot's onboard perception and the 3D model. This ensures that the simulated path-planner's coordinates map precisely to the physical vine structures in real life.

This thesis addresses this capability gap by developing a SLAM and relocalisation framework that facilitates this alignment. By synchronizing the robot's perceived position with both the physical environment and the pre-existing 3D model, this research enables the transition from static digital twins to active, autonomous robotic intervention in the field.

## 1.2 RESEARCH MOTIVATION

This research is primarily motivated by the need to enable autonomous agricultural robots to perform complex manipulation tasks, specifically robotic pruning. Unlike simple navigation, pruning requires a high-fidelity localisation and mapping (SLAM) framework that can provide the millimetre-level precision necessary for a robotic arm to interact with thin, overlapping vine structures. Developing a system that can maintain this accuracy in the presence of dynamic foliage and repetitive, unstructured row patterns is a critical bottleneck in automating viticulture and horticulture.

The demand for such technology is underscored by the economic landscape of New Zealand's primary industries. As of 2026, horticulture export revenue is forecast to reach a record $9.2 billion, with wine production constituting $2.3 billion [Industries 2025]. However, the sector faces chronic labour shortages and rising operational costs, with manual pruning alone accounting for around 23% of annual vineyard expenses [2025 Marlborough Vineyard Monitoring Report ]. The 2025 Vineyard Monitoring Report published by the Ministry for Primary Industries indicates that during the 2024/2025 season, expenditure on manual pruning increased, while spending on machine pruning declined by 21%. This shift is attributed to grower concerns regarding vine damage caused by current mechanical stripping machines [2025 Marlborough Vineyard Monitoring Report ]. These concerns highlight a critical limitation of current commercial machine-based pruning systems: insufficient precision to meet vineyard managers' operational requirements. Overcoming these precision challenges would enable robotic labour to provide a scalable response to ongoing workforce constraints, thereby supporting the long-term sustainability of New Zealand's high-value horticultural crops.

Beyond the immediate application to pruning, this work seeks to advance the research objectives of the UCVision Research Lab. By improving SLAM robustness in vegetated environments, this thesis contributes to the lab's broader mission of "seeing through occlusion" and tracking plant growth across multiple stages. The frameworks developed for semi-structured orchard conditions extend directly to other challenging environments such as forestry or search-and-rescue, providing a foundation for autonomous systems in visually complex terrain.

## 1.3 RESEARCH OBJECTIVES

The main objective of this research is to develop a precise localisation system for autonomous vineyard robots that enables accurate spatial alignment between the robot, the 3D models, and the environment.

Sub-objectives include:

- Quantify the localisation accuracy required for robotic pruning.
- Design scientific experiments and establish quantitative performance metrics to evaluate localisation accuracy.
- Achieve the necessary localisation accuracy from sub-objective 1 in vineyard environments.

- Demonstrate the practical application of the developed system by enabling precise robotic manipulation tasks such as pruning within the vineyard environment.
- Validate localisation performance under varying vineyard conditions including GNSS dropout, lighting changes, and seasonal variations.

## 1.4 THESIS STRUCTURE

This thesis is structured as follows: Chapter 2 establishes the theoretical and practical foundations of robot localisation for vineyard manipulation tasks. Chapter 3 presents the development and evaluation of a multi-sensor SLAM system for initial mapping and coarse localisation within vineyard environments. Chapter 4 describes the precise-alignment methodology, wherein robot pose estimates are refined through registration with pre-existing 3D models using stereo depth imaging, dense LiDAR, and Iterative Closest Point (ICP) algorithms. Chapter 5 concludes with a synthesis of key findings and recommendations for future research.

# Chapter 2

# BACKGROUND

## 2.1 ROBOT LOCALISATION

### 2.1.1 Introduction

Robot localisation is the fundamental process of determining the robot's pose (position and orientation) within a given frame of reference using sensory information. It is a critical step in all robot navigation and manipulation tasks which interact with the environment. Localisation is tightly linked with the domains of navigation, and mapping. Localisation and mapping are often combined in Simultaneous Localisation and Mapping (SLAM) [Ullah et al 2024] [Leonard and Durrant-Whyte 1991]. SLAM is a common precursory step for robot navigation.

This chapter discusses a historical overview of robot localisation, focusing on the foundational research that established the field. While much of this research is several decades old, the contributions in the form of mathematical formulations, algorithms, and conceptual mobile robot frameworks remain central to contemporary SLAM research. An understanding of this early research is therefore necessary in contextualising the pathways of research that have led to the state-of-the-art SLAM methods of today.

This chapter will trace the key breakthroughs that directly influence modern systems, including probabilistic localisation and mapping, multi-sensor fusion, scan matching techniques such as the Iterative Closest Point algorithm, and graph-based formulations for global consistency through pose graph optimisation. These concepts compose the theoretical and algorithmic backbone of modern SLAM, and are particularly relevant to semi-structured and uncertain environments such as vineyards and orchards.

### 2.1.2 The Inception of Mobile Robots with Reasoning

From 1966 through 1972, Stanford Research Institute (SRI) developed the world's first mobile robot, "Shakey the Robot", with the capacity to reason about its actions. It was equipped with a TV camera and an optical range finder to observe its environment. An antenna radio link communicated back to a computer, which in turn controlled the robot.

The environment was exceptionally controlled - simple enough to allow visual capabilities to be useful, but complex enough to exhibit the limits of the proposed method. The environment contained only planar surfaces. Objects were uniformly coated with red, grey or white paint

so that they were opaque with non-specular surfaces. Illumination was limited to a 2-1/2 to 1 range by using light coloured walls and floor alongside regularly spaced overhead fluorescent lights.

In the technical note on Shakey, N. J. Nilsson states that the most important result is "an appreciation of the potential complexity of the problem of vision when the real world is the subject matter" – indeed an observation that has stood the test of time well into the 21st century.

Shakey did not maintain a continuous pose estimate. Instead, localisation was formulated in discrete, semantic primitives such as "room 1", "room 2" and "door 1". The default localisation method was a form of pure open-loop, discrete dead reckoning. Shakey moved through a set of effectively atomic commands such as ROLL and TURN. The localisation system assumed the motion executed perfectly, and updated the position estimate deterministically after each command had executed. Shakey also had the ability to perform visual landmark localisation using only three primitives: doorways, convex corners and concave corners. These landmarks were pre-modelled and expected to exist in particular locations.

In an experiment where Shakey was approximately 7.5 feet away from a wall, a doorway, a convex corner and a concave corner, the robot could localise itself with an accuracy of between 5 and 10 per cent of the range, and an orientation of 5 degrees. Shakey was extremely slow, taking hours to navigate simple environments [Nilsson 1984]. Nevertheless, it set the foundation for robot localisation research over the following decades.

In 1980, Hans Moravec built upon Stanford's prior research and developed the Stanford AI Lab Cart [Moravec 1980]. For the first time, stereo vision was utilised on a mobile robot for navigation. While Shakey used multiple different sensors for perception, the Cart relied solely on visual information.

Image features, such as corners and high-variance "image patches", were represented using a multi-resolution image pyramid. Using stereo vision, the cart could determine the approximate 3D locations of these features. Similar to Shakey, the Cart was slow, moving only one metre every 10–15 minutes in discrete lurches. The cart could then derive its own motion by finding the 3D coordinate transform between subsequent observations of the same features. The 6-DOF motion matrix was estimated by solving the rigid transformation aligning the prior and current 3D feature sets using the weighted least squares method. For a 20 metre indoor course, the cart would take approximately 5 hours to navigate [Moravec 1990]. The Cart's internal map and localisation system was purely geometric, whereas Shakey, by contrast, maintained a symbolic and semantic spatial representation.

### 2.1.3 Statistical Estimation of Spatial Uncertainty

Smith and Cheeseman's 1986 research "On the Representation and Estimation of Spatial Uncertainty" [Smith and Cheeseman 1987] laid the framework for the Kalman filter, occupancy grids and other probabilistic representations of space which are still widely used today. The paper presents the spatial relationship between objects or coordinate frames as approximate transformations (ATs), which consist of an estimated pose and an associated covariance matrix

that encapsulates the spatial second-order uncertainty. The two core operations presented in this paper are *compounding* and *merging*.

Compounding is used to propagate uncertainty through sequential spatial transformations. It does this by treating the poses as random variables and propagating their uncertainty through the nonlinear composition of transformations using first-order linearization.

Let two uncertain poses be composed as

$$\mathbf{x}_3 = f(\mathbf{x}_1, \mathbf{x}_2),$$

where $\mathbf{x}_i$ denotes a pose with mean $\boldsymbol{\mu}_i$ and covariance $\mathbf{C}_i$.

The nonlinear function $f(\cdot)$ is linearized about the mean values using a first-order Taylor expansion,

$$\delta\mathbf{x}_3 \approx \mathbf{J} \begin{bmatrix} \delta\mathbf{x}_1 \\ \delta\mathbf{x}_2 \end{bmatrix},$$

where $\mathbf{J}$ is the Jacobian of $f$ evaluated at $(\boldsymbol{\mu}_1, \boldsymbol{\mu}_2)$. The covariance of the compounded transformation is then computed as

$$\mathbf{C}_3 = \mathbf{J} \begin{bmatrix} \mathbf{C}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_2 \end{bmatrix} \mathbf{J}^\top.$$

This formulation shows that uncertainty grows as transformations are compounded: variances from each step contribute additively, while correlations between position and orientation naturally emerge due to the geometry of the transformation.

Merging is used to reduce uncertainty by combining multiple independent estimates of the same spatial relationship. It does this by fusing parallel approximate transformations using a minimum-variance estimator derived from Kalman filtering.

Let two independent estimates of the same pose be given by

$$\mathbf{x}_1 \sim \mathcal{N}(\boldsymbol{\mu}_1, \mathbf{C}_1), \qquad \mathbf{x}_2 \sim \mathcal{N}(\boldsymbol{\mu}_2, \mathbf{C}_2).$$

The merged estimate $(\boldsymbol{\mu}_3, \mathbf{C}_3)$ is computed using the Kalman gain

$$\mathbf{K} = \mathbf{C}_1 \left(\mathbf{C}_1 + \mathbf{C}_2\right)^{-1},$$

with the updated mean and covariance given by

$$\boldsymbol{\mu}_3 = \boldsymbol{\mu}_1 + \mathbf{K}\left(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1\right),$$

$$\mathbf{C}_3 = \mathbf{C}_1 - \mathbf{K}\mathbf{C}_1.$$

This operation produces a weighted average of the two estimates, where estimates with lower covariance have greater influence. As a result, the merged covariance is smaller than either input covariance, reflecting the reduction in uncertainty obtained by fusing independent information sources.

Together, these operations allow uncertainty to be consistently propagated and reduced within a network of spatial relations, referred to as a *relational map*.

The foundational paper for Kalman filtering was produced by Rudolf E. Kalman in 1960, which introduced the concept of recursive state prediction and uncertainty propagation in the time domain. However, it was not until Smith and Cheeseman's paper titled "Estimating Uncertain Spatial Relationships in Robotics" that the Extended Kalman Filter was applied to the SLAM problem. They introduced the concept of the "Stochastic Map", which aggregated the estimated sensor poses and estimated landmark poses into a single, unified vector. The covariance matrix describes the deeply correlated relationship between uncertainty measurements in sensor and landmark poses. This concept marked a departure in robot localisation research from deterministic methods to probabilistic methods.

Building on the work from Smith and Cheeseman [Smith and Cheeseman 1987], Alberto Elfes from Carnegie Mellon University made the observation that previous geometric approaches to robot perception lacked the ability to handle sensor uncertainty and errors, confining the applicability to highly structured domains [Elfes 1989]. For more complex and unstructured environments, robot perception systems had to address unexpected circumstances and handle environments with little or no prior knowledge of the terrain. Thus, in 1989, Elfes introduces occupancy grids for robot perception. An occupancy grid is a tessellated representation of spatial information, constituted by a multi-dimensional lattice of cells that represent free space. The occupancy grid maintains stochastic estimates of the occupancy state of each cell within the lattice. The occupancy estimates are derived from probabilistic sensor models of incoming range sensor data [Elfes 1989].

Mapping is therefore constructed by incrementally integrating sensor measurements and estimating the occupancy of regions of space within a probabilistic spatial representation, rather than explicitly extracting geometric features such as in the Stanford Cart [Moravec 1990] or using a symbolic spatial representation like Shakey the robot [Nilsson 1984]. A proposed framework is illustrated in fig. 2.1, whereby multiple sensor views are composed into a local sensor map, and then multiple sensor maps are fused into a single global map with a consistent reference frame. While the exact methods of integrating and fusing sensor data have developed immensely over the past 30 years, this framework remains the backbone of many modern SLAM techniques today.

fig. 2.2 illustrates a sonar mapping robot utilising occupancy grids for navigation. While this robot was still "stop-and-go", unable to integrate sensor measurements during motion, similar to Shakey the robot [Nilsson 1984] and the Cart [Moravec 1990], the resolution of the global map was sufficient to distinguish walls and open doors.

In fig. 2.3, Elfes shows an example of fusing sonar and scanline stereo maps. Both sensor datasets are initially converted into the common occupancy grid format before being superimposed. The stereo excels in angular resolution; however, unmarked surfaces or low-contrast edges are not detected well, and uncertainty scales with distance. Sonar detects surfaces well but has poor angular resolution due to the large beam width. Comparatively, sonar has low range uncertainty.

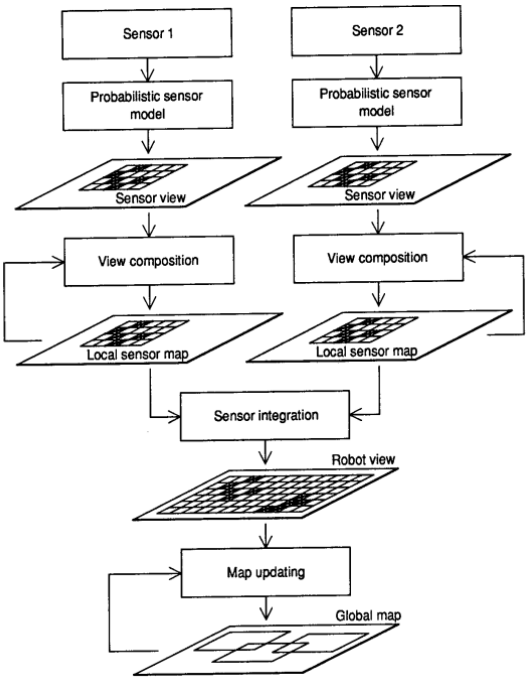



Figure 2.1: A framework for occupancy-grid-based robot mapping. [Elfes 1989]

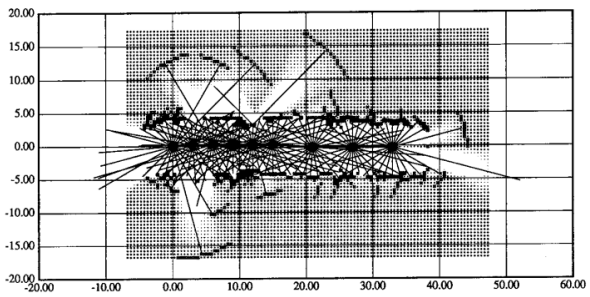



Figure 2.2: Sonar mapping along a corridor [Elfes 1989].

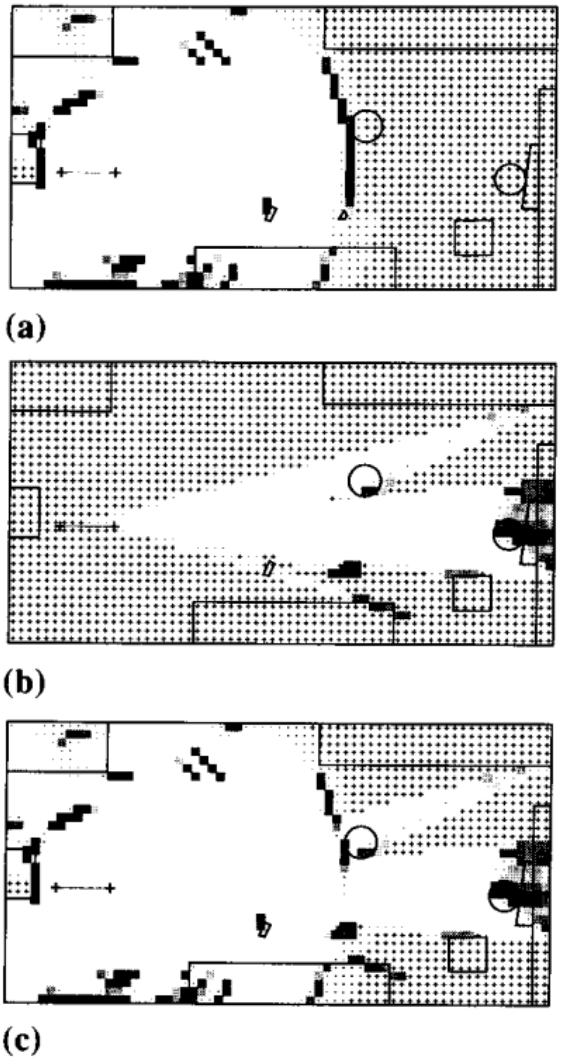



Figure 2.3: Sensor integration of sonar and scanline stereo. Occupancy grids generated separately for sonar (a) and scanline stereo (b), as well as the integrated map (c) obtained through sensor fusion, are shown. Occupied regions are marked by shaded squares, empty areas by dots fading to white space, and unknown spaces by + marks [Elfes 1989].

Elfes asserts that in order to merge multiple sensor views, acquired by the robot at different positions, into a coherent world map, one must be able to accurately estimate the robot's motion. This statement summarises the co-dependence between localisation and mapping that persists in modern-day SLAM. Elfes notes that dead reckoning estimates are not sufficient, and that positional error is compounded over sequences of movement – as mathematically formulated in Smith and Cheeseman's aforementioned compounding operation [Smith and Cheeseman 1987]. This compounding uncertainty necessitates the ability to explicitly track positional uncertainty and take it into account when composing a world map.

Elfes employs the approximate transformation (AT) framework proposed by Smith, Self and Cheeseman [Smith and Cheeseman 1987]. The robot motion is initially derived from dead reckoning and inertial navigation. For more precise positional estimation, a multi-resolution correlation-based motion solver is utilised by searching for the optimal registration between the current robot view and the global map. Low resolution results are propagated up as priors for subsequent higher resolution registrations. Uncertainty is propagated over time by applying a convolving operation around the robot position, effectively "blurring" the occupancy grid. As the robot position accumulates error over time, older positions are "sharp" while newer positional estimates become increasingly "fuzzy", offering less and less useful information. Eventually, the robot's position "dissolves" in a "cloud of probabilistic smoke" [Elfes 1989]. Unlike purely geometric maps, the architecture proposed by Elfes explicitly handles the uncertainty from a multi-sensor composition of information in an integrated framework. This framework simplified the localisation problem fundamentally, and paved the way for the development of Simultaneous Localisation and Mapping (SLAM) in the 1990s and beyond.

### 2.1.4 Optimisation Methods for Pose Estimation

The 1992 paper titled "A Method for Registration of 3-D Shapes" by Besl and McKay [Besl and McKay 1992] introduces the *Iterative Closest Point (ICP)* algorithm, an algorithm for rigid 3D registration, which later became the foundation of real-time SLAM and specifically LiDAR-based SLAM. The paper frames 3-D registration as an optimisation problem in which the rigid 3D transformation between two point sets is estimated by minimising the error between corresponding points. The ICP algorithm proceeds iteratively by alternating between finding new correspondences using a closest-point criterion and executing closed-form least-squares estimation of the optimal rigid-body transformation between correspondence sets. The paper provides a proof that demonstrates that the algorithm converges monotonically to a local minimum with respect to the mean-square distance objective function. However, convergence is sensitive to initial conditions and partial overlap – while a local minimum is guaranteed, it by no means guarantees the global minimum.

This foundational work was built on by Yang Chen and Gerard Medioni in their paper "Object Modeling by Registration of Multiple Range Images" [Chen and Medioni 1991] which introduced the point-to-plane error metric, which greatly improved the robustness for environments with smooth surfaces, such as what one would find in and around buildings. Further research into the efficiency of the algorithm produced papers such as "Efficient variants of the

ICP algorithm" by Rusinkiewicz et al. [Rusinkiewicz and Levoy 2001]. This research was critical in making ICP efficient enough for real-time SLAM. In 2009, Segal et al. introduced the "Generalised-ICP" [Segal et al 2009] which combined point-to-point, point-to-plane, and plane-to-plane into a generalised algorithm which significantly improved robustness in unstructured environments.

Lu & Milios introduced pose graph optimisation [Lu and Milios 1997], which offered a mathematical optimisation-based alternative to the filtering approaches, such as Kalman filters, information filters and particle filters, which dominated the SLAM landscape. Filtering techniques like EKF struggled with scalability and linearisation in large environments as they essentially "forgot" past states after they are integrated into the global model. This approach results in an inconsistent world model as different parts of the model are updated independently [Lu and Milios 1997]. This is in comparison with optimisation techniques such as pose graph optimisation which maintain the entire history of poses and systematically propagate pose optimisations to all related frames.

Sensor data measured at a single robot pose is called a "data frame". Spatial relations between frames are derived from scan-matching or from odometry measurements. In the pose graph, local frame positions (i.e. robot poses) are stored as variables, and spatial relations between frames are stored as constraints. In traditional graph terminology, the variables are "nodes" and the constraints are "edges". The objective of pose graph optimisation is to optimise the estimates of the variables using the network of the constraints. Through this process, data frames can be aligned and registered to create a coherent global map [Lu and Milios 1997].

## 2.2 ROBOT LOCALISATION IN VINEYARD AND ORCHARD ROBOTICS

Unlike structured industrial settings, agricultural environments are "semi-structured" environments characterised by homogeneous structures, seasonal variability, and dynamic occlusion [Hroob et al 2021]. Moreover, the reliance on Global Navigation Satellite Systems (GNSS) often falls short in orchards as dense canopies can attenuate satellite signals or induce multipath errors, making standard GNSS insufficient for centimetre-level precision tasks [Tang et al 2024]. Therefore, research has focused on developing technologies that use semantic understanding, geometric constraints, and multi-sensor fusion to operate reliably in these GNSS-denied environments [Tang et al 2024].

### 2.2.1 Environmental Constraints and Challenges

Semi-structured agricultural environments create unique constraints that pose a challenge for generic SLAM algorithms designed for structured urban environments.

**Perceptual Aliasing and Geometric Repetitiveness** Commercial crops in vineyards and orchards are planted in long, parallel rows with uniform spacing to maximise yield and facilitate machine access. The long uniform rows create a "corridor effect", especially for LiDAR-based

systems. The corridor effect is characterised by the absence of distinct geometric features along the longitudinal position of the robot (the direction of movement down the row), such that the longitudinal axis becomes unobservable, which can lead to significant drift in the estimated trajectory as the robot cannot determine if it has travelled one metre forward or five metres forward [Khan et al 2025]. Visual systems face a similar challenge wherein standard feature descriptors (e.g., ORB, SIFT) extracted from one part of a row are statistically indistinguishable from those in another, resulting in erroneous feature matching and false-positive loop closure [Papadimitriou et al 2022].

**Dynamic and Unstructured Environment** Many SLAM systems make an assumption of a static world, however orchards and vineyards are dynamic environments. Short-term dynamics such as wind-induced canopy motion, workers, and machines can pose challenges to mapping and loop closure. Long-term dynamics include seasonal changes, wherein a map constructed during the dormant winter season can differ greatly compared with the full-canopy summer season. This can pose challenges to long-term localisation and map maintenance [Schmidt et al 2024].

Furthermore, the uneven terrain can cause camera shake and wheel slippage that introduces high-frequency noise into odometry estimates [Zhang et al 2026].

### 2.2.2 LiDAR SLAM

LiDAR SLAM is often favoured due to its illumination invariance, but standard scan-matching algorithms like ICP often fail in symmetric vineyard rows due to the aforementioned perceptual aliasing along the longitudinal axis.

VineSLAM tries to address the perceptual aliasing problem by grouping point clouds into specific vineyard features: semi-planes (for the continuous canopy/trellis) and points (for trunks/posts). It utilises a Particle Filter that fuses these features; semi-planes offer lateral constraints to keep the robot centred, and point features provide longitudinal constraints along the row. Comparative studies show VineSLAM outperforms general-purpose algorithms like LeGO-LOAM in symmetric corridors [Aguiar et al 2022].

Tree-SLAM detects individual tree trunks within the point clouds, mapping the environment as a sparse graph of tree landmarks rather than a dense point cloud. This method achieved localisation errors as low as 18 cm for individual trees (less than 20% of standard planting distance) in apple and pear orchards [Rapado-Rincon and Kootstra 2025]. This object-level approach improves navigation but also facilitates agronomic tasks such as per-tree yield monitoring [Rapado-Rincon and Kootstra 2025].

The April-LIO-SAM framework tightly couples LiDAR-Inertial Odometry (LIO-SAM) with AprilTags acting as artificial landmarks. April tags provide absolute pose constraints that constrain the drift from LiDAR odometry. Experimental validation in orchards demonstrated that with tags spaced at 3 to 5 metres, the system achieves a root-mean-square error (RMSE) of approximately 0.057 m, maintaining stability even under occlusion and varying illumination [Guan et al 2025]. Another multi-sensor system achieved positioning errors of 0.57 cm in pear orchards, using visual data for loop closure and LiDAR for geometric structure [Tang et al 2024].

**Benchmarking Datasets** Progress in the field is accelerated by open-source datasets tailored to agricultural challenges:

- **MAgro Dataset:** Collected in apple and pear orchards using LiDAR, stereo cameras, and RTK-GPS. It is designed to benchmark loop closure and localisation under varying weather and illumination conditions [Schmidt et al 2024] [Marzoa Tanco et al 2024].
- **Bacchus Long-Term (BLT) Dataset:** A unique dataset capturing a vineyard over a 7-month period. It supports research into "lifelong SLAM," challenging algorithms to handle the drastic morphological changes of the canopy from winter dormancy to summer harvest [Polvara et al 2024].
- **AgriLiRa4D:** A recent multi-modal dataset (LiDAR, Camera, 4D Radar) collected via UAV over diverse terrains (flat, hilly), facilitating research into radar-based fusion and aerial coverage paths [Zhan et al 2026].

### 2.2.3 Conclusion

The development of SLAM for orchards and vineyards has evolved from the application of generic algorithms to the design of domain-specific architectures. Current research prioritises semantic SLAM to filter dynamic agricultural noise and object-level mapping (e.g., Tree-SLAM) to resolve geometric ambiguity. Furthermore, the integration of multi-sensor fusion (e.g., April-LIO-SAM) has proven essential for achieving the centimetre-level accuracy required for precision tasks. Future work is increasingly driven by long-term autonomy, leveraging datasets like BLT to enable robots that can adapt to the seasonal biological evolution of the crop [Tang et al 2024] [Polvara et al 2024].

## 2.3 ROBOT LOCALISATION IN VINEYARD PRUNING

Pruning typically occurs in the dormant season (winter), which introduces a unique robot localisation challenge in the form of geometric sparsity and the lack of distinctive visual features caused by the absence of foliage [Aguiar et al 2022]. The environment consists of thin, bare branches that provide a low density of LiDAR points and few distinctive visual textures for cameras to "lock" onto [Ahmed et al 2023]. Furthermore, without leaves to block the view, sensors often pick up distant "noise" from the sky or neighbouring rows, leading to errors in depth estimation and foreground segmentation [You et al 2022]. The following analysis details the strategies used for spatial alignment with vines, specifically in the context of robotic pruning platforms.

### 2.3.1 Spatial Alignment Modalities for Pruning Vines

While traditional passive vision remains the baseline for high-resolution 3D modelling, it faces significant geometric and environmental hurdles. To address the ill-conditioned configurations of binocular systems – specifically when linear canes align with the stereo baseline – Botterill et

al. employ a trinocular rig consisting of three 1.3-megapixel cameras arranged in a triangular configuration [Botterill et al 2017]. This allows for the dynamic selection of the mathematically optimal camera pair for specific cane segments, a level of geometric robustness that passive binocular systems lack.

However, even optimised trinocular systems remain vulnerable to outdoor lighting variability. This necessitates a shift toward active sensing, either through illumination or direct range-finding. Systems like "Bumblebee" integrate active light stereo cameras to generate dual point clouds, which are then refined via Iterative Closest Point (ICP) algorithms to maintain invariance against ambient shadows [Navone et al 2025, Silwal et al 2021]. In contrast, Felicetti et al. bypass the lighting and correspondence problem entirely by utilising downward-facing 3D LiDAR [Felicetti et al 2025]. By applying a minimalist depth-thresholding algorithm within a specific range ($D_{min}$ to $D_{max}$, typically 250–500 mm), this approach sacrifices the fine-grained topological "expert system" understanding provided by Botterill's trinocular rig in favour of raw speed, processing a vine in just 7.3 seconds [Felicetti et al 2025, Botterill et al 2017].

The computational backend of these systems further distinguishes their operational philosophies. Botterill et al. prioritise spatial stability for robotic manipulation through a heavy incremental bundle adjustment pipeline. By utilising the *g2o* graph optimisation framework and robust Pseudo-Huber cost functions to penalise large residuals, they achieve trajectory errors below 1%, though at a higher computational cost. Recent paradigms, however, suggest that such global optimisation may be unnecessary for effective pruning. Navone et al. and the "Archie Jnr" platform utilise deep panoptic segmentation models (e.g., YOLOv8-seg or Detectron2) to extract semantic coordinates directly in image space [Navone et al 2025, Hizatate et al 2025].

By registering these masks with depth data, these neural architectures achieve an $mAP_{50-95}$ of 87.2% for pruning point identification, effectively bridging the gap between the high-fidelity reconstruction of Botterill et al. and the high-speed obstacle-navigation of Felicetti et al. [Botterill et al 2017, Felicetti et al 2025, Navone et al 2025]. Ultimately, while the trinocular approach provides the most detailed 3D map for precise spur pruning, the integration of active illumination and semantic segmentation offers a more computationally viable path toward commercial field speeds.

Table 2.1: Comparison of Perception Modalities for Vineyard Localisation

| **System** | **Modality** | **Spatial Accuracy** | **Speed/Throughput** |
|---|---|---|---|
| Botterill et al. [Botterill et al 2017] | Trinocular Stereo | High (Millimetre) | 120 s/vine |
| Silwal et al. [Silwal et al 2021] | Active Stereo | High (Millimetre) | 137 s/vine |
| Felicetti et al. [Felicetti et al 2025] | 3D LiDAR | Moderate (Depth-filter) | 7.3 s/vine |
| Archie Jnr [Navone et al 2025] | RGB-D (YOLOv8) | High (3D Coordinates) | 15.2 s/cut point |

# Chapter 3

# SLAM

## 3.1 INTRODUCTION

This chapter focuses on the development and implementation of a SLAM system designed to operate independently of frequent GNSS updates. The primary objective is to ensure high-fidelity localisation in GNSS-denied or GNSS-degraded scenarios. Furthermore, a critical requirement for this system is temporal robustness; that is, the SLAM front-end must remain invariant to fluctuating visual appearances, enabling continuous 24/7 operation across both diurnal and nocturnal cycles, different weather conditions, and plant growth cycles. Thirdly, as aforementioned, vineyard and orchard rows form long, symmetric, corridor-like structures which limits the geometric diversity available to scan-matching algorithms, making them susceptible to ambiguity and longitudinal drift. The system presented in this chapter is therefore not intended as a general-purpose SLAM solution; it is a specialised architecture targeted at the structural and perceptual characteristics of row-crop environments such as vineyards and orchards.

### 3.1.1 Motivation

The necessity for this research stems from the operational limitations of the current UCVision Rover localisation system. Prior iterations of this platform utilised an Extended Kalman Filter (EKF) heavily reliant on Real-Time Kinematic (RTK) GPS data. While precise under open skies, this dependency proved non-viable for robust field operations where tree canopies, topography, and public RTK base station blackouts frequently cause signal multipath or dropout.

Transitioning to a LiDAR-centric SLAM approach, tailored to these agricultural environments, provides three distinct advantages:

- **GNSS-independent localisation:** By shifting the primary motion estimation to scan-matching and loop closure, the system becomes less reliant on RTK.
- **Corridor robustness:** The architecture presented here specifically addresses the structural challenges of row-crop environments through vineyard-focused parameterisation and sensor fusion suited to row-structured terrain.
- **Downstream utility:** Beyond localisation, the high-density LiDAR maps generated can be used for additional tasks such as landmark detection (e.g., vine trunks and trellis posts),

spatial alignment of Gaussian splats captured from opposing sides of a row, and global registration of data across multiple vineyard bays.

## 3.2 RELATED WORK

The development of Simultaneous Localisation and Mapping (SLAM) has shifted from laboratory settings to complex, real-world deployments across autonomous driving, aerial vehicles, and subterranean exploration. Due to the limitations of external positioning systems such as GNSS in dynamic or urban environments, Light Detection and Ranging (LiDAR) has become the dominant sensing modality, providing high-fidelity, active 3D geometry independent of ambient illumination. The state of the art in this field is defined by LiDAR-Inertial Odometry (LIO), which tightly couples LiDAR measurements with high-frequency inertial data from an Inertial Measurement Unit (IMU) to compensate for motion distortion and maintain observability in geometrically degenerate scenes [Koide et al 2024]. Modern LIO research is primarily distinguished by a bifurcation of techniques into two dominant optimisation philosophies: Error-State Kalman Filtering (ESKF) and Factor Graph Optimisation (FGO). This review traces the architectural evolution of LIO, comparing and contrasting leading algorithms such as LIO-SAM, FAST-LIO2, GLIM, and Super-LIO to synthesise the current trends in computational efficiency and global consistency.

### 3.2.1 Modern LiDAR-SLAM Architectures

The foundation of modern LiDAR SLAM rests on the seminal LOAM (LiDAR Odometry and Mapping) framework [Zhang and Singh 2014], which introduced the critical concept of extracting and registering geometric features, specifically classifying points into "edge" and "planar" surfaces based on curvature. LOAM's split-process architecture separated high-frequency odometry from low-frequency mapping. Subsequent work, such as LeGO-LOAM [Shan and Englot 2018], refined this by optimizing ground points separately. However, these systems were loosely-coupled and feature-dependent, leading to inevitable drift over long trajectories and fragility in unstructured environments (e.g., foliage or feature-poor corridors).

The transition to tight-coupling was facilitated by the advancement of IMU Pre-integration theory [Forster et al 2017]. This re-parameterization allows IMU integration to be expressed as a relative motion constraint (e.g., $\Delta\mathbf{R}_{ij}$) that is independent of the initial velocity and pose estimate. This innovation decouples the high IMU rate from the optimization rate, making it computationally feasible for graph-based methods to efficiently optimize state biases and velocities over large time windows.

The result of this evolution is LIO-SAM [Shan et al 2020], which established the feature-based factor graph benchmark for global consistency. LIO-SAM builds a factor graph incorporating LiDAR odometry, IMU pre-integration, and explicit loop closure factors. To manage computational complexity, it employs a "sliding window" local map for scan-to-map matching. Despite its accuracy and robustness compared to predecessors, its reliance on LOAM-style

feature extraction and a standard KD-tree for neighbor searching limits its performance in geometrically degenerate environments.

The current state of the art is defined by two architectural tracks: those prioritising extreme computational efficiency via filtering, and those prioritising global consistency and robustness via dense graph optimisation.

**Filter-Based Direct Odometry** Algorithms in the embedded track are designed to minimize CPU usage, making them suitable for low-power ARM processors on platforms like small drones. These systems rely on the Error-State Kalman Filter (ESKF) and are primarily odometry engines that estimate state and covariance, but are susceptible to drift over time as past states are marginalized.

FAST-LIO2 [Xu et al 2022] is a widely adopted benchmark for computational efficiency, eschewing feature extraction in favour of registering raw points directly. Its two principal innovations are the Incremental KD-Tree (iKD-Tree), which supports dynamic map updates with amortised $O(\log N)$ complexity, and a Kalman Gain Inversion that reduces the filter update cost from the measurement dimension to the 18-dimensional state vector, enabling kHz-rate processing on modest hardware. Point-LIO [He et al 2023] extends this paradigm by updating the EKF state for every individual point as it arrives from the sensor, yielding a processing bandwidth bounded only by the sensor's sampling rate. This point-wise update strategy enables tracking under aggressive motion where conventional scan-accumulation methods fail due to intra-frame distortion. Super-LIO [Wang et al 2025] represents the most recent advance in this track, targeting the map data structure rather than the estimator itself. It introduces the **OctVox** representation — a hybrid combining a spatial hash map with internal octrees — to enforce strict point density constraints, and pairs this with a Heuristic-Guided $k$-Nearest Neighbour (HKNN) search that further reduces per-scan processing time. Reported benchmarks indicate lower CPU utilisation than FAST-LIO2, suggesting OctVox-based representations may define the next generation of resource-constrained LIO.

**Graph-Based Global Consistency** Systems in this track prioritize long-term mapping quality and robustness by leveraging high-performance hardware, particularly embedded GPUs.

GLIM (Global LiDAR Inertial Mapping) [Koide et al 2024] represents a distinct departure from CPU-centric design, instead leveraging GPU parallelism to achieve dense, globally consistent mapping. Scan matching in GLIM is implemented via GPU-accelerated Voxelised Generalised Iterative Closest Point (VGICP) factors, which allow the system to incorporate nearly all points per frame and thereby retain geometric constraints that feature-based methods routinely discard. A further key contribution is the Submap Endpoint Mechanism, in which IMU pre-integration factors are inserted as stiff constraints on unobservable degrees of freedom within each submap. This structural guarantee preserves registration quality even when the LiDAR geometry is degenerate — for example in featureless tunnels or elevator shafts. For back-end estimation, GLIM employs a fixed-lag smoother that continuously optimises a bounded window of recent poses, combining the accuracy benefits of iterative re-linearisation with bounded memory consumption, in contrast to full-history SLAM approaches.

This trade-off between computational efficiency and global consistency is summarized in the comparison table below. The most significant recent innovations have occurred in the map data structure domain (iKD-Tree, OctVox) for the filter-based track, and in hardware-software co-design (GLIM's GPU acceleration) for the graph-based track.

Table 3.1: Comparative Analysis of State-of-the-Art LIO Systems

| **Algorithm** | **Backend** | **Map Structure** | **Loop Closure** | **Robustness (Degeneracy)** | **Efficiency (CPU)** |
|---|---|---|---|---|---|
| LIO-SAM | Factor Graph | Static KD-Tree | Yes | Moderate | Low |
| FAST-LIO2 | Iterated ESKF | iKD-Tree | No | High (Direct) | High |
| Point-LIO | Iterated ESKF | iKD-Tree | No | Very High | Moderate |
| DLIO | Geometric Obs. | KD-Tree/iVox | No | High | High |
| GLIM | Factor Graph | GPU Voxelmap | Yes | **Extreme** | Low (GPU) |
| Super-LIO | Iterated ESKF | **OctVox** | No | High | **Very High** |

Super-LIO is effective for projects constrained to low-power CPUs that can tolerate bounded drift, while GLIM excels at high-fidelity mapping applications where long-term global consistency and robustness to severe degeneracy are paramount, provided a GPU accelerator is available.

### 3.2.2 SLAM Hyperparameter Optimisation

The optimisation of SLAM systems has evolved from manual tuning into rigorous hyperparameter optimisation (HPO) and statistical space exploration. As robots transition from the lab into industrial, agricultural and extraterrestrial domains, stability, robustness and generalisation become pre-eminent [Wei et al 2026]. Many modern SLAM systems are configured by a complex set of interdependent parameters that influence behaviour from high-level feature extraction to back-end pose graph optimisation [Han et al 2020]. The challenge of configuring a high-dimensional space, often referred to as the curse of dimensionality, poses a significant hurdle in tuning these SLAM systems. The first challenge is finding a subset of points from a high-dimensional search space that accurately represents the underlying performance landscape [Han et al 2020].

**Search Space Sampling** One such method for sampling high-dimensional search spaces is Latin Hypercube Sampling (LHS). LHS, as formally introduced by Michael McKay in 1979, was used as a variance reduction technique to address the inefficiencies of Monte Carlo simulation (MCS) in high-dimensional domains [Chrisman 2014]. In MCS, sample points are generated independently, which often leads to "clusters" and "voids" in disparate regions, which can neglect critical narrow regions of optimal performance [Deutsch and Deutsch 2012]. LHS addresses this by enforcing a stratification of the marginal distributions of each input variable.

LHS divides the range of each of the $d$ variables into $n$ equiprobable intervals. In a standard uniform distribution of $[0, 1]$, these strata are defined as $A_i = \left(\frac{i-1}{n}, \frac{i}{n}\right]$ for $i = 1, \ldots, n$

[Chrisman 2014]. To generate a LHS sample, one value is drawn from each interval for each dimension. These values are then combined into vectors such that each $n$ sample point is the only representative in each axis-aligned hyperplane. A comparison of LHS, MCS and Grid Search is detailed in Table 3.2.

| **Sampling Attribute** | **Monte Carlo (MCS)** | **Grid Search** | **Latin Hypercube (LHS)** |
|---|---|---|---|
| **Variable Coverage** | Random; susceptible to clustering | Uniform but sparse for high $d$ | Guaranteed univariate uniformity [Chrisman 2014] |
| **Sample Efficiency** | Low; requires large $n$ for stability | Very low; $n^d$ scaling | High; independent of $d$ [Deutsch and Deutsch 2012] |
| **Bias/Variance** | Unbiased; high variance | Biased to grid points | Consistent; asymptotically unbiased [Packham and Schmidt 2008] |
| **Search Structure** | Unstructured | Rigidly structured | Near-random with low discrepancy [Chrisman 2014] |

Table 3.2: Comparison of Sampling Strategies

The advantage of LHS in SLAM parameterisation comes from its capacity to represent the parameter space with fewer samples than traditional methods. In a SLAM system with ten parameters, a grid search with only three levels per parameter would require $3^{10} = 59,049$ evaluations. Testing each sample on a real trajectory quickly becomes time prohibitive. LHS can explore the same search space with fewer points while maintaining that each parameter is tested at different levels [Deutsch and Deutsch 2012]. However, while uniformity is guaranteed across each axis, it does not guarantee that the interior of the hypercube is filled uniformly. As a result, researchers have developed Latin Hypercube Sampling with Multidimensional Uniformity (LHSMDU) to address the limitations of univariate stratification [Deutsch and Deutsch 2012]. LHSMDU employs a sequential elimination process, where a large pool of MCS candidates are generated, and those that are too close in the multidimensional space are pruned [Deutsch and Deutsch 2012]. This method maximises the minimum inter-point distance, maximising the spread of samples throughout the multi-dimensional space.

Often the SLAM parameters have physical or co-dependent constraints, such as turn radii or computational resource limits. Standard LHS assumes independent variables, but extensions like Latin Hypercube Sampling with Dependence (LHSD) allow for the incorporation of copula-based dependence structures [Packham and Schmidt 2008]. This method enforces constraints on the sampling strategy so that the search does not enter undefined/invalid regions of the search space.

**Robustness, Generalisation and Overfitting** Often, it is not difficult to find a high-performance SLAM configuration for a single dataset. The real challenge is finding a config-

uration that generalises across a diverse set of environments. This is the common problem in machine learning of overfitting to a dataset, and is a widespread bottleneck to reliable and robust deployment of robotic systems [Soares et al 2025].

Different SLAM architectures are more sensitive to overfitting to specific environmental characteristics. For example, visual SLAM systems are more sensitive to overfitting to different lighting conditions, and LiDAR SLAM systems are more sensitive to overfitting to specific geometric characteristics. For example, a LiDAR SLAM parameter set tuned for a feature-rich, static hallway will likely adopt aggressive feature matching thresholds and high confidence in loop closure. When transitioned to an open field with sparse landmarks and faster movement, these same parameters may cause the system to lose tracking or generate false-positive loop closures [soumyadip 2024]. Some general examples of environmental factors, SLAM failure modes and impacts on tuning are illustrated in Table 3.3. This highlights the importance of avoiding overfitting to a single dataset, as characteristics such as "low light" for VSLAM can skew the optimisation process in a direction that is unsuitable for other lighting conditions.

| **Environmental Factor** | **VSLAM Failure Mode** | **LiDAR SLAM Failure Mode** | **Impact on Tuning** |
|---|---|---|---|
| **Low Light** | Feature extraction failure [10] | Minimal (Active sensing) | Requires adaptive thresholds [10] |
| **Dynamic Objects** | Tracking drift; ghosting [Soares et al 2025] | Segmentation errors [soumyadip 2024] | Necessitates robust loss kernels [Soares et al 2025] |
| **Low Texture** | Tracking loss [19] | Geometric degeneracy [soumyadip 2024] | High dependency on sensor fusion [Wei et al 2026] |
| **High Speed** | Motion blur [Xu et al 2024] | Point cloud skew [soumyadip 2024] | Requires higher frame rates [Xu et al 2024] |

Table 3.3: Impact of Environmental Factors on SLAM Performance

Traditional optimisation focuses on minimising a scalar error metric, such as Absolute Trajectory Error (ATE), on a single "representative" dataset – typically one of the prevailing SLAM benchmark datasets such as KITTI or TUM RGB-D. However, this approach tends to "chase" the noise of that specific trajectory [Kandasamy et al ], resulting in parameters that are essentially "hard-coded" to that specific sequence of sensor observations and noise realisations of the training data.

## 3.3 PROPOSED METHODS

The following subsections describe the methods developed and integrated to achieve globally consistent, ENU-referenced localisation in vineyard environments. These contributions build on the GLIM framework surveyed in the preceding section and address the key challenges of reference frame alignment, GNSS outlier rejection, and parameter optimisation.

To begin this investigation, we implemented GLIM. We chose GLIM as it had infrastructure in place to add extension modules to incorporate new sensors and capabilities. GLIM provides a global callback slot mechanism that allows access to the internal states of the mapping process and insertion of additional constraints into the factor graph. GLIM also eliminated sensor-specific processes, meaning it can be applied to a wide variety of range sensors such as spinning-type LiDAR, non-repetitive scan LiDAR, solid-state LiDAR and even RGB-D cameras.

GLIM state estimation is created in the local odometry frame. For our use in the wider research project, we required state estimation in the local ENU reference frame, which is a surveyed-in point at each of our farms. Thus, we needed a procedure to align the GLIM reference frame with the existing local ENU reference frame.

Firstly, we time synchronise the on-board GPS unit with the LiDAR and with the system clock. This allows us to associate corresponding GNSS points with the submaps or keyframes of LiDAR frames that GLIM registers. We aggregate a set of corresponding GNSS coordinates with keyframe coordinates, and then implement the Kabsch algorithm to find the relative transform between the two point sets by minimising the root-mean-square deviation between the two point sets.

The Kabsch algorithm relies on the least-squares method to minimise the root-mean-square deviation (RMSD) between two point sets. Since the objective function squares the distances between paired points, outliers produce disproportionately large influences on the final transformation matrix. Thus, for global alignment, it is important that outliers are filtered out before executing the Kabsch algorithm.

### 3.3.1 Geometric Consistency-Based Outlier Rejection

To remove spurious GNSS measurements prior to initialisation, we apply a geometric consistency check that exploits the rigidity of the GNSS–IMU transformation. The method assumes that, for inlier measurements, pairwise distances between GNSS observations should be preserved when mapped into the local map frame via the estimated poses.

Let $\{\mathbf{g}_i\}_{i=1}^N$ denote GNSS positions expressed in the ENU frame, and $\{\hat{\mathbf{g}}_i\}_{i=1}^N$ denote the corresponding GNSS positions expressed in the map frame, obtained by transforming IMU poses with a fixed extrinsic calibration. For each point $i$, we evaluate its consistency with all other points $j \neq i$ by comparing pairwise Euclidean distances in both frames as shown in Equation (3.1):

$$|\|\mathbf{g}_i - \mathbf{g}_j\| - \|\hat{\mathbf{g}}_i - \hat{\mathbf{g}}_j\|| < \tau, \tag{3.1}$$

where $\tau$ is a distance consistency threshold reflecting accepted SLAM error and GNSS noise.

This approach was chosen over stochastic methods like RANSAC or iterative M-estimators because it takes advantage of the invariance of pairwise Euclidean distances under rigid body transformations, which means that the method remains entirely deterministic and independent of any initial pose estimate or probabilistic modelling.

To formalise the inlier classification, we define a pairwise consistency indicator $c_{ij}$ in Equa-

tion (3.2):

$$c_{ij} = \begin{cases} 1 & \text{if } |\|\mathbf{g}_i - \mathbf{g}_j\| - \|\hat{\mathbf{g}}_i - \hat{\mathbf{g}}_j\|| < \tau \\ 0 & \text{otherwise} \end{cases} \tag{3.2}$$

A measurement is classified as an inlier if the fraction of other points with which it satisfies the above constraint exceeds a minimum inlier ratio $\rho$, satisfying the condition:

$$\frac{1}{N-1} \sum_{j \neq i}^{N} c_{ij} > \rho. \tag{3.3}$$

GNSS multipath errors can often create erroneous clusters of points that are mutually consistent, and therefore are not filtered out by conventional statistical outlier methods. However, the majority-based criterion in Equation (3.3) ensures that points are only kept if they belong to the global rigid consensus, effectively filtering out clusters of mutually consistent but globally erroneous points. After classification, only inlier GNSS measurements and their corresponding map-frame estimates are preserved for subsequent processing.

### 3.3.2 Planar Geometry Quality Assessment in 2D

Subsequently to outlier rejection, we assess the geometric quality of the remaining point set under the assumption of planar motion by analysing its distribution in the horizontal plane. This analysis is motivated by the sensitivity of the Kabsch algorithm to degenerate point configurations: near-collinear or poorly distributed trajectories produce ill-conditioned cross-covariance matrices, leading to unstable or biased transformations. By detecting such configurations before alignment, we ensure the SVD is well-conditioned and that the estimated transformation is numerically reliable. In particular, we enforce minimum requirements on translational baseline ($\geq 5.0\,\text{m}$), planar spread, rotational excitation, and numerical conditioning.

Given a set of 3D points $\{\mathbf{p}_i\}_{i=1}^{N}$, only the horizontal components are considered by projecting each point onto the $xy$-plane, yielding $\mathbf{q}_i = [x_i, y_i]^\top$. If fewer than three points are available, the geometry is deemed insufficient and the assessment terminates early.

The projected points are assembled into a data matrix $\mathbf{M}$ as defined in Equation (3.4):

$$\mathbf{M} = \begin{bmatrix} \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_N \end{bmatrix} \in \mathbb{R}^{2\times N}, \tag{3.4}$$

which is centred by subtracting the mean $\bar{\mathbf{q}}$ from each column. Singular value decomposition is then applied to the centred matrix as shown in Equation (3.5):

$$\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top, \tag{3.5}$$

yielding two singular values $\sigma_1 \geq \sigma_2$ that characterise the principal spatial extents of the trajectory in the plane.

The condition number $\kappa$ is defined in Equation (3.6):

$$\kappa = \frac{\sigma_1}{\sigma_2}, \tag{3.6}$$

and is used as a degeneracy metric, with large values indicating near-collinear motion. The geometry is considered non-degenerate if $\kappa < 20.0$.

To evaluate spatial coverage, a 2D spread ratio $r$ is computed according to Equation (3.7):

$$r = \frac{\sigma_2}{\sigma_1}, \tag{3.7}$$

which measures how well the trajectory excites both planar directions. A minimum spread ratio of $r \geq 0.15$ is required. Additionally, a minimum displacement constraint is enforced by requiring both singular values to exceed a baseline threshold of 5.0 m. The spatial distribution is considered sufficient only if both the spread ratio and displacement conditions are satisfied.

Rotational excitation is evaluated independently by computing the angular span of the point set in the horizontal plane. The geometry is considered to provide sufficient rotational constraint if the angular span exceeds 45.0°.

The final geometry quality assessment combines these criteria, reporting whether the point set exhibits sufficient planar spread, adequate rotational excitation, and non-degenerate conditioning. This 2D-based evaluation provides a robust and numerically stable measure of geometric suitability for planar motion scenarios.

### 3.3.3 GLIM Reference Frame Alignment with Local ENU

Given two corresponding point sets $\{\mathbf{a}_i\}_{i=1}^N, \{\mathbf{b}_i\}_{i=1}^N \subset \mathbb{R}^3$, the goal is to estimate a rigid-body transformation $(\mathbf{R}, \mathbf{t}) \in \mathrm{SO}(3) \times \mathbb{R}^3$ that minimises the sum of squared alignment errors as defined in the following objective function:

$$\min_{\mathbf{R},\mathbf{t}} \sum_{i=1}^{N} \|\mathbf{b}_i - (\mathbf{R}\mathbf{a}_i + \mathbf{t})\|^2, \quad \text{s.t. } \det(\mathbf{R}) = 1. \tag{3.8}$$

First, the centroids of both point sets are computed as:

$$\boldsymbol{\mu}_A = \frac{1}{N}\sum_{i=1}^{N} \mathbf{a}_i, \qquad \boldsymbol{\mu}_B = \frac{1}{N}\sum_{i=1}^{N} \mathbf{b}_i. \tag{3.9}$$

The point sets are then centred by subtracting their respective centroids:

$$\tilde{\mathbf{a}}_i = \mathbf{a}_i - \boldsymbol{\mu}_A, \qquad \tilde{\mathbf{b}}_i = \mathbf{b}_i - \boldsymbol{\mu}_B. \tag{3.10}$$

A $3 \times 3$ cross-covariance matrix $\mathbf{H}$ is formed as:

$$\mathbf{H} = \sum_{i=1}^{N} \tilde{\mathbf{b}}_i \tilde{\mathbf{a}}_i^\top. \tag{3.11}$$

The singular value decomposition (SVD) of $\mathbf{H}$ is computed in Equation (3.12):

$$\mathbf{H} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top. \tag{3.12}$$

To ensure a proper rotation and avoid reflections, a diagonal correction matrix $\mathbf{D}$ is introduced:

$$\mathbf{D} = \mathrm{diag}(1,\ 1,\ \mathrm{sign}(\det(\mathbf{U}\mathbf{V}^\top))). \tag{3.13}$$

The optimal rotation $\mathbf{R}$ is then given by:

$$\mathbf{R} = \mathbf{U}\mathbf{D}\mathbf{V}^\top. \tag{3.14}$$

Finally, the translation $\mathbf{t}$ is recovered according to Equation (3.15):

$$\mathbf{t} = \boldsymbol{\mu}_B - \mathbf{R}\boldsymbol{\mu}_A. \tag{3.15}$$

The resulting rigid transformation $(\mathbf{R}, \mathbf{t})$ aligns the source point set to the target point set in a least-squares sense. This static transformation is published to the ROS2 transform tree to link the local ENU frame to the GLIM mapping frame.

### 3.3.4 GNSS Prior Factor Construction

This procedure augments an existing SLAM factor graph with GNSS-based prior constraints that relate submap poses to global position measurements. Each GNSS measurement is incorporated as a probabilistic constraint on the estimated submap pose through a known sensor extrinsic calibration.

Let $\mathbf{T}_{\text{map}}^{\text{enu}} \in SE(3)$ denote the rigid transformation from the local ENU frame to the GLIM map frame, and $\mathbf{p}_{\text{imu}}^{\text{gps}} \in \mathbb{R}^3$ the fixed translation from the IMU frame to the GNSS antenna frame. For each correspondence between a submap and a GNSS measurement, the following steps are performed.

First, the GNSS position $\mathbf{p}_{\text{enu}} \in \mathbb{R}^3$ is extracted in the ENU frame and transformed into the map frame as shown in Equation (3.16):

$$\mathbf{p}_{\text{map}} = \mathbf{T}_{\text{map}}^{\text{enu}}\, \mathbf{p}_{\text{enu}}. \tag{3.16}$$

Next, a diagonal Gaussian noise model is constructed from the reported GNSS covariance. Let $\boldsymbol{\sigma} \in \mathbb{R}^3$ denote the standard deviations obtained from the square root of the covariance diagonal. These values are scaled by a configurable factor $\alpha$ and converted to a covariance matrix in Equation (3.17):

$$\boldsymbol{\Sigma}_{\text{gps}} = \mathrm{diag}\big((\alpha\boldsymbol{\sigma})^2\big)\,. \tag{3.17}$$

Given a submap pose $\mathbf{T}_{\text{map}}^{\text{sub}} \in SE(3)$, the predicted GNSS antenna position in the map frame is expressed in Equation (3.18):

$$\hat{\mathbf{p}}_{\text{map}} = \mathbf{T}_{\text{map}}^{\text{sub}}\, \mathbf{p}_{\text{imu}}^{\text{gps}}. \tag{3.18}$$

This formulation is valid as the GLIM submap origin is defined at the IMU center, such that the extrinsic offset $\mathbf{p}_{\text{imu}}^{\text{gps}}$ represents the antenna position relative to the submap origin. An

expression-based factor is then constructed to minimize the residual between the predicted and measured positions under the noise model $\mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_{\text{gps}})$:

$$\mathbf{r}(\mathbf{T}_{\text{map}}^{\text{sub}}) = \|\hat{\mathbf{p}}_{\text{map}} - \mathbf{p}_{\text{map}}\|_{\mathbf{\Sigma}_{\text{gps}}}^2 . \tag{3.19}$$

This factor directly constrains the corresponding submap pose variable in the factor graph, contributing to global consistency and LiDAR-inertial odometry drift correction. In effect, this factor constrains the GLIM-predicted position of the GPS antenna using the measured GNSS position at that time.

### 3.3.5 Robust SLAM Parameterisation via Iterative Region Refinement

The optimisation of Simultaneous Localisation and Mapping (SLAM) parameters is a high-dimensional, non-convex problem where the objective is to identify a parameter vector $\Theta$ that minimises localisation error across heterogeneous environments [Han et al 2020]. This subsection proposes a model-free, heuristic region-refinement algorithm that transitions from global stochastic exploration to localised deterministic refinement. The strategy assumes that optimal configurations cluster within specific sub-regions of the hyper-dimensional search space $\Omega$, which are iteratively identified and contracted using statistical distribution analysis.

#### 3.3.5.1 Problem Formulation

We define the SLAM system as a complex function $f(X, \theta)$, where $X$ represents raw sensor data (e.g., LiDAR scans and inertial measurements) and $\Theta$ is the $d$-dimensional parameter vector. In a typical LiDAR-inertial setup, $\Theta$ includes parameters such as LiDAR resolution and registration strategy. The objective is to find $\Theta^*$ such that:

$$\Theta^* = \arg\min_{\Theta \in \Omega} \mathcal{J}(\Theta) \tag{3.20}$$

where $\mathcal{J}$ is a robust objective function representing the consensus performance across $K$ distinct motion profiles.

#### 3.3.5.2 Sampling via Latin Hypercube Strategy

To explore the initial parameter manifold, we utilise Latin Hypercube Sampling (LHS) as an alternative to traditional grid search or independent Monte Carlo (MC) sampling [Deutsch and Deutsch 2012]. LHS is technically categorised as a stratified sampling design [Chrisman 2014]. For a sample size $N$ in $d$ dimensions, the range of each parameter $\theta_j$ is partitioned into $N$ equiprobable intervals for $j = 1, \ldots, d$ [Chrisman 2014]. The mathematical constraint — ensuring exactly one sample per row and column of the grid — minimises the "clustering" phenomenon common in pure random sampling [Deutsch and Deutsch 2012]. This stratification ensures that the marginal distribution of each parameter is sampled uniformly, achieving a lower variance than independent Monte Carlo sampling for the same sample size [Packham and Schmidt 2008].

**Implementation and Normalisation** Parameter vectors are first normalised to the unit hypercube $[0,1]^d$ [Deutsch and Deutsch 2012]. To ensure reproducibility, the pseudo-random permutations used to construct the hypercube are seeded deterministically. The samples are subsequently scaled to physical ranges using:

$$\theta_{j,i} = \theta_{j,\min} + u_{j,i}(\theta_{j,\max} - \theta_{j,\min}) \tag{3.21}$$

where $\theta_{j,\min}$ and $\theta_{j,\max}$ represent the physical lower and upper bounds for parameter $j$.

#### 3.3.5.3 The Multi-Dataset Consensus Framework

A primary challenge in SLAM optimisation is the risk of "over-specialisation", where parameters become overfitted to the unique noise characteristics of a single trajectory [Soares et al 2025]. We implement a Consensus Survival Framework to prioritise reliability across datasets.

In some situations, standard "Mean-Performance Optimisation" can hide brittle configurations that perform well on average but fail catastrophically on a single sequence [Kandasamy et al ]. However, in this setting, catastrophic failure typically results in excessively high amplitude errors, which effectively inflate the mean normalised error. That being said, some particular parameter configurations can lead to excessive GPU draw, resulting in a runtime CUDA out-of-memory error. If the parameter set does not complete processing the trajectory on any dataset $k$, it is assigned an infinite cost and discarded from the survivor pool. This ensures the survival of only those configurations that meet the computational requirements of the system.

**Formal Objective Function** For the "surviving" configurations, the objective function $J(\theta)$ is defined as the mean normalized error across the valid sequences (3.22):

$$J(\theta) = \begin{cases} \frac{1}{K}\sum_{k=1}^{K} E_k(\theta) & \text{if success} \in \{D_1, \ldots, D_K\} \\ \infty & \text{otherwise} \end{cases} \tag{3.22}$$

where $E_k(\theta)$ is the primary performance metric for dataset $k$.

#### 3.3.5.4 Iterative Region Refinement (IRR)

The optimisation proceeds through an "Iterative Contraction" or "Funnel" strategy. This mirrors the reduction of a search radius in traditional Trust Region Methods, concentrating the sampling density into the most promising basins of attraction.

**Statistical Search Space Contraction** At the conclusion of each phase $i$, the top-performing configurations are used to calculate new search boundaries for phase $i+1$. We employ the 10th and 90th percentiles of the elite set as robust estimators for the update rule in Equation (3.23):

$$\Omega_{j,i+1} = [P_{10}(\theta_{j,\text{top}}), P_{90}(\theta_{j,\text{top}})]. \tag{3.23}$$

This quantile-based reduction is superior to using absolute Min/Max values because it is resilient to "lucky" outliers — configurations that may perform well due to noise interactions rather than parameter stability.

**Refinement Phase Schedule** The optimisation is structured into four phases, increasing the selection pressure (lower survival rate) as the search space contracts, as detailed in table 3.4:

Table 3.4: Evolutionary Phases of the Sampling Algorithm

| Phase | Objective | Iterations (N) | Survival Rate | Description |
|---|---|---|---|---|
| Phase 1 | Thinning | 100 | 30% | Broad exploration identifying stable regions. |
| Phase 2 | Narrowing | 100 | 20% | Convergence on local minima. |
| Phase 3 | Refinement | 100 | 10% | High-density sampling within basins. |
| Phase 4 | Fine-tuning | 75 | 1.3% | Final tuning to find top candidate. |

This incremental strategy is distinct from Successive Halving (SHA), as it preserves a continuous distribution of potential solutions rather than discarding them in discrete early-stopping steps, which is critical for SLAM where early performance does not always correlate with final drift.

#### 3.3.5.5 Optimisation Metrics

For the optimisation of the base GLIM algorithm, the cost function for candidate ranking is the Trajectory Error Percentage ($TEP$), calculated as the percentage of the ratio of Absolute Trajectory Error ($ATE$) to the total trajectory length ($L$):

$$TEP = \frac{ATE}{L} \times 100\%. \tag{3.24}$$

The $TEP$ effectively measures the per cent drift per metre of motion.

For the optimisation of the proposed method, the cost function is a custom metric referred to as Global Alignment and Smoothness Profile ($GASP$). This is calculated by taking the weighted sum of the $ATE$ per metre and the standard deviation of the translational Relative Pose Error ($RPE$), as defined in Equation (3.25):

$$\text{GASP} = w_{\text{drift}} \cdot \left(\frac{ATE}{L}\right) + w_{\text{smoothness}} \cdot \text{Translational RPE}. \tag{3.25}$$

The weights $w_{\text{drift}}$ and $w_{\text{smoothness}}$ have been empirically tuned such that each term contributes approximately equally to the total cost. Because the weighted sum mixes quantities with different units, GASP serves as a relative performance index; its values are meaningful only when compared within the same experimental framework.

## 3.4 RESULTS

This section presents the empirical results from the evaluation of three localisation configurations: the baseline GLIM algorithm with default parameters, the refined GLIM algorithm with optimised parameters, and the proposed GNSS-augmented method. Each configuration was evaluated on five vineyard datasets. Performance is reported using Root Mean Square Error (RMSE), Absolute Trajectory Error (ATE), Relative Pose Error (RPE), and, for the proposed method, the Global Alignment and Smoothness Profile (GASP) metric defined in the methodology.

### 3.4.1 Baseline GLIM Performance

table 3.5 reports the performance of the baseline GLIM algorithm across the five datasets. The error percentage per metre ranges from 1.253% (Dataset 1) to 7.361% (Dataset 2). The 3D RMSE ranges from 8.035 m (Dataset 1) to 38.107 m (Dataset 3), and the ATE ranges from 6.605 m to 30.444 m. The maximum single-point 3D error observed is 80.339 m in Dataset 2.

Table 3.5: Baseline GLIM performance over the five test datasets

| **Metrics** | **Dataset 0** | **Dataset 1** | **Dataset 2** | **Dataset 3** | **Dataset 4** |
|---|---|---|---|---|---|
| RMSE x (m) | 20.776 | 7.253 | 4.097 | 4.747 | 3.598 |
| RMSE y (m) | 3.730 | 2.766 | 3.391 | 4.596 | 4.889 |
| RMSE z (m) | 15.107 | 1.747 | 35.970 | 36.908 | 25.879 |
| RMSE 3D (m) | 26.990 | 8.035 | 36.755 | 38.107 | 27.767 |
| ATE (m) | 19.877 | 6.605 | 25.921 | 30.444 | 25.331 |
| max error 3D (m) | 47.892 | 17.766 | 80.339 | 69.878 | 56.729 |
| error per meter (m) | 0.0340 | 0.0130 | 0.0740 | 0.0490 | 0.0420 |
| error percent per meter (%) | 3.3580 | 1.2530 | 7.3610 | 4.8970 | 4.1910 |
| RPE 1m translation RMSE | 0.4697 | 0.2944 | 0.7632 | 1.0127 | 0.5356 |
| RPE 1m translation mean | 0.3100 | 0.1390 | 0.3210 | 0.3790 | 0.3170 |
| RPE 1m translation std | 0.4125 | 0.2602 | 0.6767 | 0.9384 | 0.4285 |
| RPE 1m translation max | 3.4500 | 2.8930 | 4.1800 | 15.7180 | 2.8810 |
| RPE 10m translation RMSE | 2.3790 | 1.5040 | 3.3400 | 3.3230 | 2.5920 |
| RPE 10m translation mean | 2.2680 | 1.1230 | 2.8110 | 3.0490 | 2.4040 |
| RPE 10m translation std | 0.7134 | 1.0076 | 1.8240 | 1.4120 | 0.9390 |
| RPE 10m translation max | 6.6530 | 7.9020 | 11.2070 | 16.2210 | 7.0210 |

### 3.4.2 Refined GLIM Performance

table 3.6 reports the results for the refined GLIM algorithm following parameter optimisation. The error percentage per metre ranges from 0.6081% (Dataset 1) to 3.6150% (Dataset 2). The 3D RMSE ranges from 3.4808 m (Dataset 1) to 25.1789 m (Dataset 2), and the ATE ranges from 3.2315 m to 12.6781 m.

Table 3.6: Refined GLIM performance over the five datasets

| **Metrics** | **Dataset 0** | **Dataset 1** | **Dataset 2** | **Dataset 3** | **Dataset 4** |
|---|---|---|---|---|---|
| RMSE x (m) | 6.3487 | 2.5059 | 3.6146 | 3.5669 | 1.1221 |
| RMSE y (m) | 3.2186 | 2.3727 | 24.9149 | 3.4499 | 2.6844 |
| RMSE z (m) | 0.6987 | 0.4545 | 0.3994 | 1.0413 | 2.1387 |
| RMSE 3D (m) | 7.1522 | 3.4808 | 25.1789 | 5.0705 | 3.6110 |
| ATE (m) | 5.9190 | 3.2315 | 12.6781 | 4.6763 | 3.3144 |
| max error 3D (m) | 14.3770 | 5.5867 | 71.0506 | 13.0417 | 5.8597 |
| error per meter (m) | 0.0087 | 0.0061 | 0.0362 | 0.0075 | 0.0067 |
| error percent per meter (%) | 0.8735 | 0.6081 | 3.6150 | 0.7499 | 0.6680 |
| RPE 1m translation RMSE | 0.3907 | 0.2609 | 0.5692 | 0.8317 | 0.3689 |
| RPE 1m translation mean | 0.1471 | 0.0899 | 0.3037 | 0.1133 | 0.1358 |
| RPE 1m translation std | 0.3619 | 0.2449 | 0.4814 | 0.8239 | 0.3430 |
| RPE 1m translation max | 3.1190 | 2.8235 | 2.3976 | 15.1426 | 2.7154 |
| RPE 10m translation RMSE | 1.2280 | 1.5267 | 4.6931 | 2.6912 | 1.4290 |
| RPE 10m translation mean | 0.7309 | 0.5967 | 2.6938 | 0.8912 | 0.5826 |
| RPE 10m translation std | 0.9868 | 1.4052 | 3.8430 | 2.5394 | 1.3048 |
| RPE 10m translation max | 5.7503 | 7.7271 | 10.7189 | 15.4780 | 6.6573 |

### 3.4.3 Proposed Method Performance

table 3.7 reports the results for the proposed GNSS-augmented method. The error percentage per metre ranges from 0.0153% (Dataset 4) to 0.0501% (Dataset 2). The 3D RMSE ranges from 0.1097 m (Dataset 0) to 0.2280 m (Dataset 2), and the ATE ranges from 0.0811 m to 0.1741 m. The GASP metric, which combines drift rate and local pose smoothness, returned values between 0.0274 (Dataset 0) and 0.0679 (Dataset 2).

Table 3.7: Proposed method performance over the 5 test datasets

| **Metrics** | **Dataset 0** | **Dataset 1** | **Dataset 2** | **Dataset 3** | **Dataset 4** |
|---|---|---|---|---|---|
| RMSE x (m) | 0.0756 | 0.0528 | 0.0909 | 0.0528 | 0.0728 |
| RMSE y (m) | 0.0645 | 0.0906 | 0.2003 | 0.0906 | 0.0789 |
| RMSE z (m) | 0.0465 | 0.0347 | 0.0599 | 0.0347 | 0.0560 |
| RMSE 3D (m) | 0.1097 | 0.1105 | 0.2280 | 0.1105 | 0.1211 |
| ATE (m) | 0.0846 | 0.0811 | 0.1741 | 0.0811 | 0.1031 |
| max error 3D (m) | 0.7363 | 0.8963 | 1.0246 | 0.8963 | 0.7056 |
| error per meter (m) | 0.0002 | 0.0002 | 0.0005 | 0.0002 | 0.0002 |
| error percent per meter (%) | 0.0171 | 0.0154 | 0.0501 | 0.0154 | 0.0153 |
| gasp | 0.0274 | 0.0290 | 0.0679 | 0.0290 | 0.0429 |
| RPE 1m translation RMSE | 0.0270 | 0.0308 | 0.0556 | 0.0308 | 0.0444 |
| RPE 1m translation mean | 0.0175 | 0.0206 | 0.0282 | 0.0206 | 0.0249 |
| RPE 1m translation std | 0.0205 | 0.0229 | 0.0479 | 0.0229 | 0.0368 |
| RPE 1m translation max | 0.2001 | 0.4906 | 0.6992 | 0.4906 | 0.6462 |
| RPE 10m translation RMSE | 0.0649 | 0.0936 | 0.1772 | 0.0936 | 0.0769 |
| RPE 10m translation mean | 0.0433 | 0.0613 | 0.0987 | 0.0613 | 0.0573 |
| RPE 10m translation std | 0.0482 | 0.0708 | 0.1471 | 0.0708 | 0.0512 |
| RPE 10m translation max | 0.3799 | 0.7620 | 0.8566 | 0.7620 | 0.5377 |

### 3.4.4 Comparative Summary

table 3.8 presents a side-by-side comparison of the three methods on three key metrics: error percentage per metre, RPE 1 m translation standard deviation, and error per metre. The lowest value for each metric–dataset combination is indicated in bold.

Table 3.8: Comparison Table of the Key Metrics (Best results in bold)

| Metric | Method | Dataset 0 | Dataset 1 | Dataset 2 | Dataset 3 | Dataset 4 |
|---|---|---|---|---|---|---|
| error percent per meter (%) | Baseline GLIM | 3.3580 | 1.2530 | 7.3610 | 4.8970 | 4.1910 |
| | Refined GLIM | 0.8735 | 0.6081 | 3.6150 | 0.7499 | 0.6680 |
| | Proposed Method | **0.0171** | **0.0154** | **0.0501** | **0.0154** | **0.0153** |
| RPE 1m translation std | Baseline GLIM | 0.4125 | 0.2602 | 0.6767 | 0.9384 | 0.4285 |
| | Refined GLIM | 0.3619 | 0.2449 | 0.4814 | 0.8239 | 0.3430 |
| | Proposed Method | **0.0205** | **0.0229** | **0.0479** | **0.0229** | **0.0368** |
| error per meter (m) | Baseline GLIM | 0.0340 | 0.0130 | 0.0740 | 0.0490 | 0.0420 |
| | Refined GLIM | 0.0087 | 0.0061 | 0.0362 | 0.0075 | 0.0067 |
| | Proposed Method | **0.0002** | **0.0002** | **0.0005** | **0.0002** | **0.0002** |

### 3.4.5 GNSS Factor Utilisation

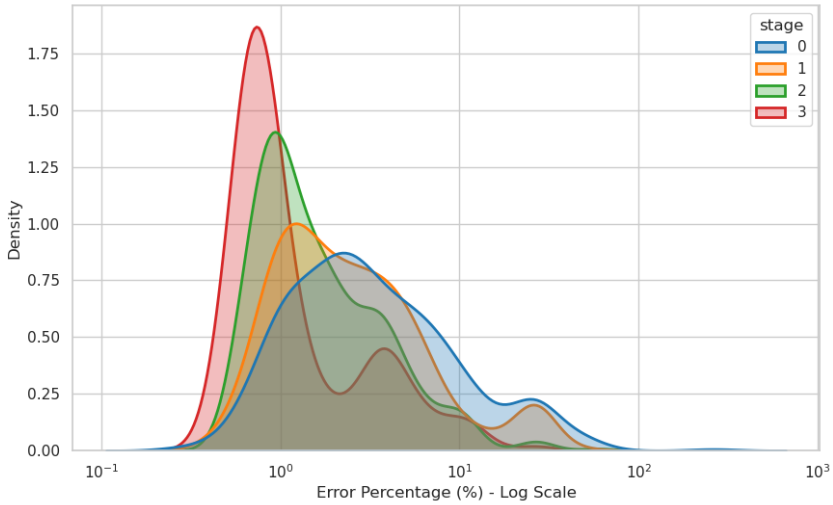
table 3.9 reports the number of GNSS factors incorporated into the proposed method's factor graph relative to the total number of available GNSS measurements. Across the five datasets, the proportion of GNSS points retained after outlier rejection and incorporated as factors ranges from 0.65% (Dataset 2) to 0.78% (Dataset 3).

Table 3.9: Statistics on the number of GNSS factors used for the proposed method

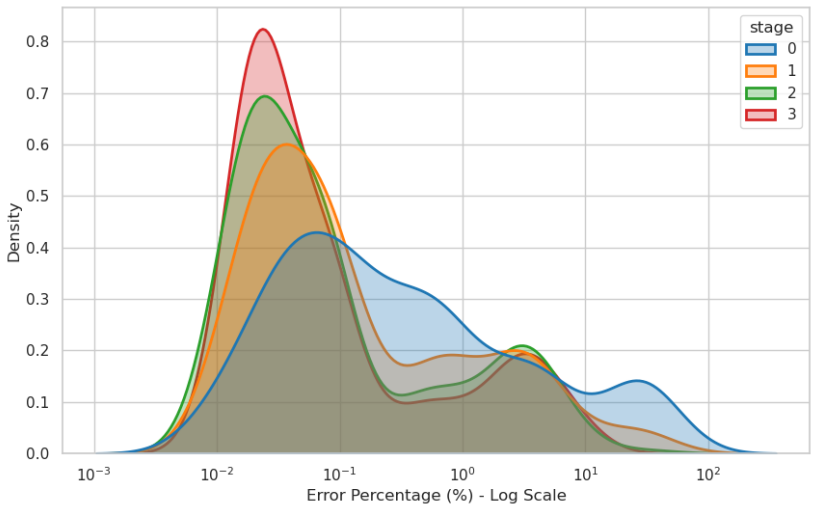
| Metric | Dataset 0 | Dataset 1 | Dataset 2 | Dataset 3 | Dataset 4 |
|---|---|---|---|---|---|
| Number of factors | 93 | 73 | 47 | 86 | 69 |
| Number of GNSS points | 13461 | 10090 | 7236 | 11071 | 9241 |
| Percent of GNSS points used | 0.69 | 0.72 | 0.65 | 0.78 | 0.75 |

### 3.4.6 Parameter Optimisation Progression

fig. 3.1 shows the distribution of per-iteration errors across the four optimisation stages. fig. 3.1a plots the error percentage per metre for the baseline GLIM tuning, and fig. 3.1b plots the GASP metric for the proposed method tuning. Both plots use a logarithmic horizontal axis.



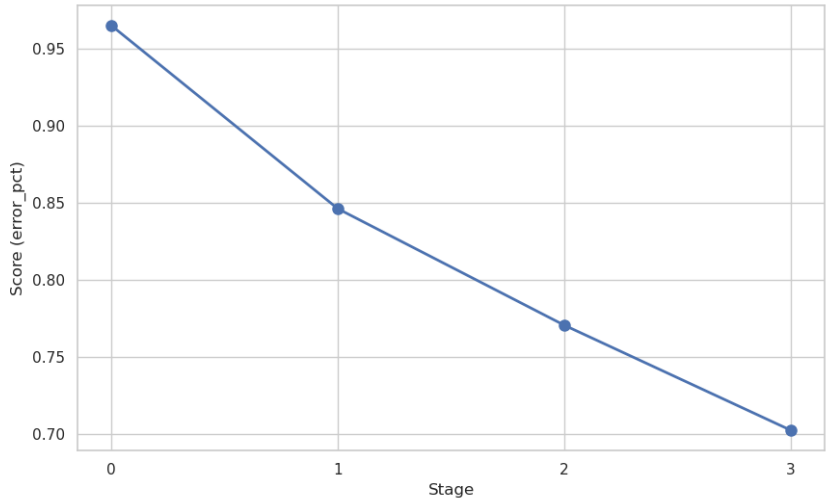

(a) Baseline glim tuning.



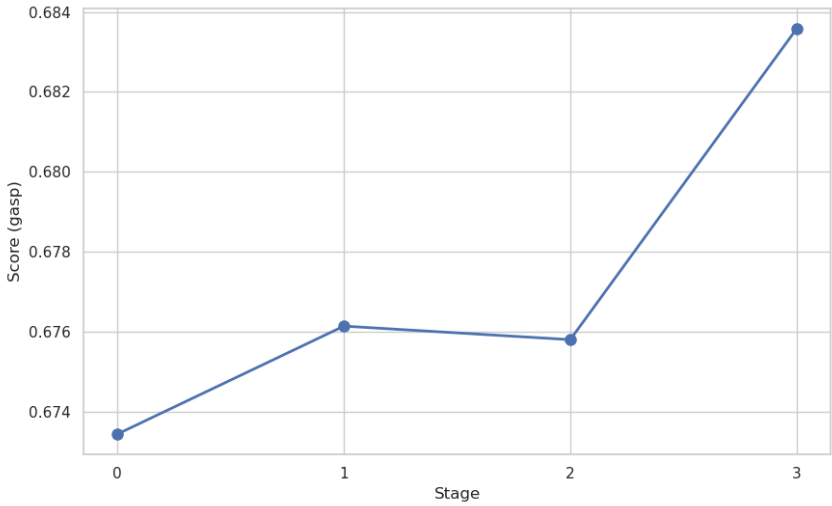

(b) Proposed method tuning.

Figure 3.1: Error distribution (log scale) over optimisation stages for baseline tuning (error = error percentage) and proposed method tuning (error = GASP).

fig. 3.2 plots the best consensus error at the end of each optimisation stage. fig. 3.2a tracks the error percentage for the baseline GLIM, and fig. 3.2b tracks the GASP metric for the proposed method. The baseline error decreases monotonically across the four stages; the proposed method's GASP score increases from stage 0 to stage 3.




(a) Baseline GLIM error percentage.

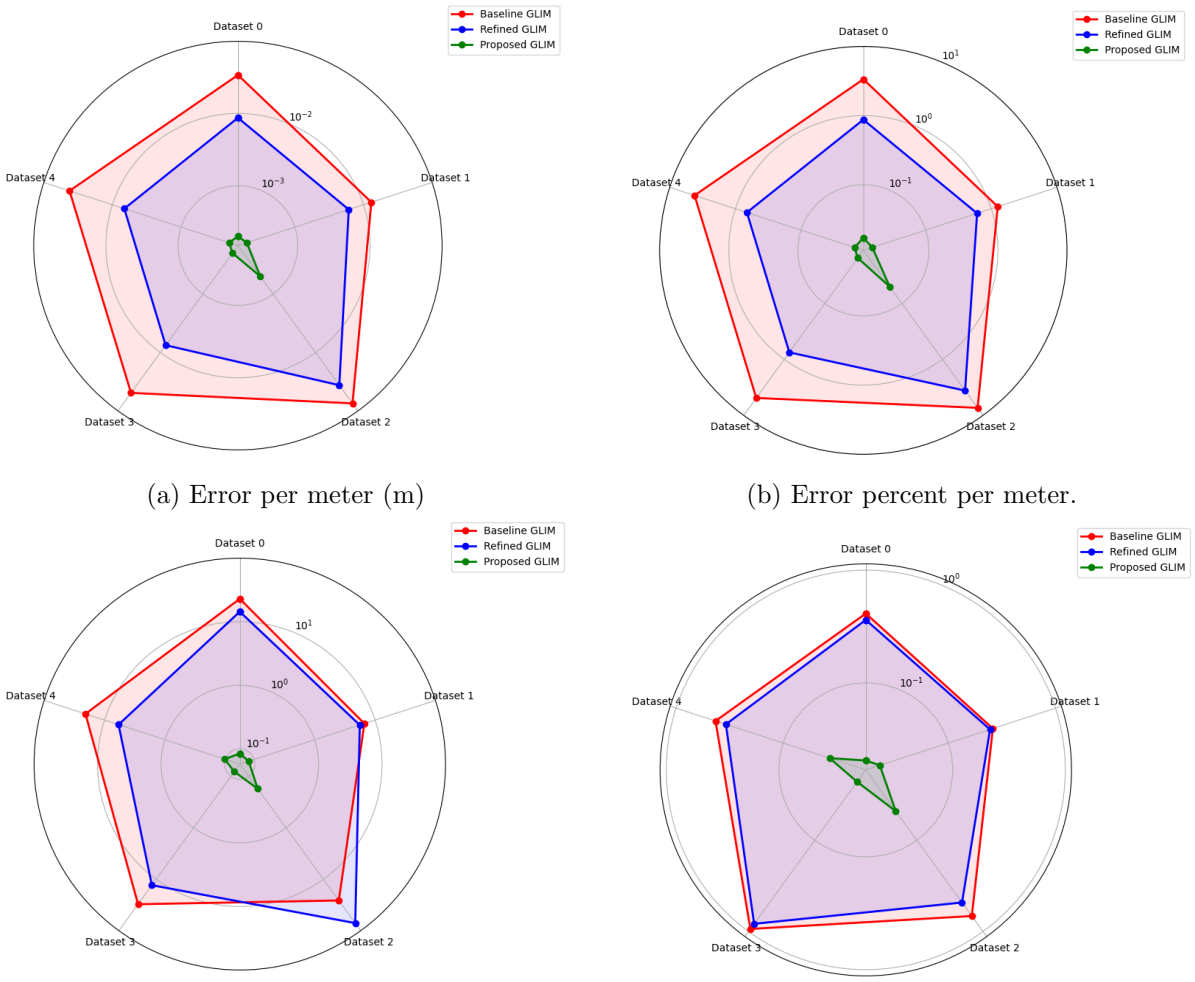



(b) Proposed Method GASP error.

Figure 3.2: Consensus error over optimisation stages for baseline tuning (error = error percentage) and proposed method tuning (error = GASP).

### 3.4.7 Multi-Metric Radar Comparison

fig. 3.3 presents four radar plots comparing the baseline, refined, and proposed methods across all five datasets. The four metrics displayed are error per metre, error percentage per metre, ATE, and RPE 1 m translation standard deviation. Radial axes are plotted on a logarithmic scale.




(a) Error per meter (m)

(b) Error percent per meter.

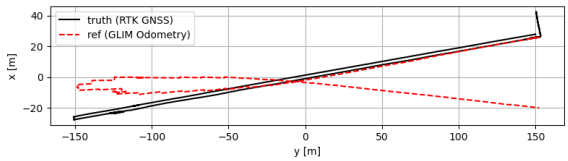
(c) ATE error (m).

(d) RPE (translation only) standard deviation.

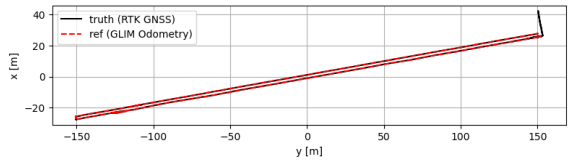
Figure 3.3: Spider plot comparisons of the glim baseline, refined and proposed methods for various key performance metrics.

### 3.4.8 Trajectory Visualisation

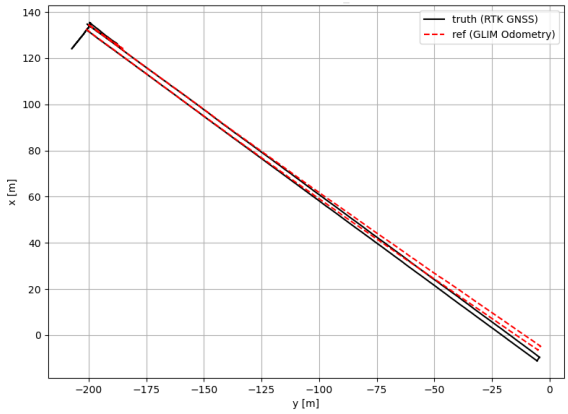
fig. 3.4 presents the planar ($x$–$y$) trajectories for each dataset. The left column shows the refined GLIM output and the right column shows the proposed method output, both plotted against the RTK GNSS ground truth.



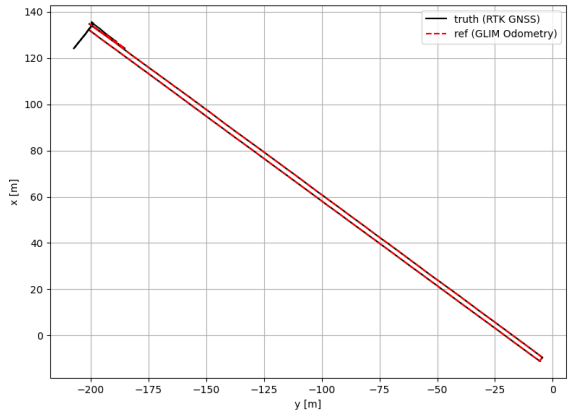

(a) Baseline — Dataset 0



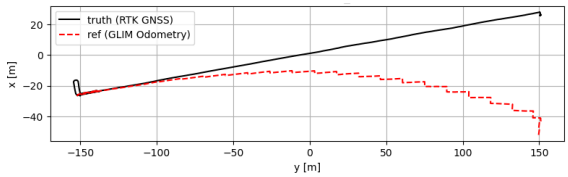

(b) Proposed — Dataset 0



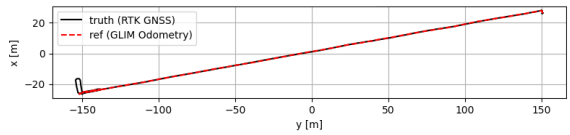

(c) Baseline — Dataset 1



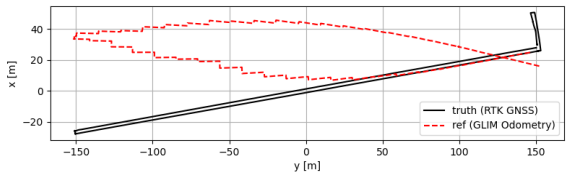

(d) Proposed — Dataset 1



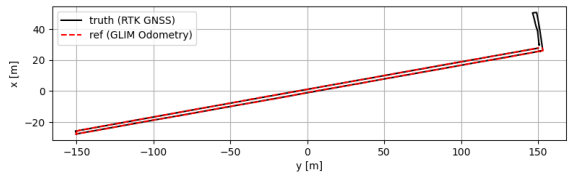

(e) Baseline — Dataset 2



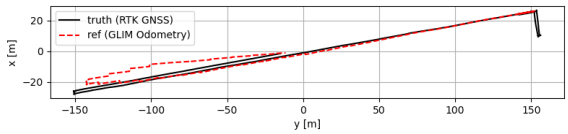

(f) Proposed — Dataset 2



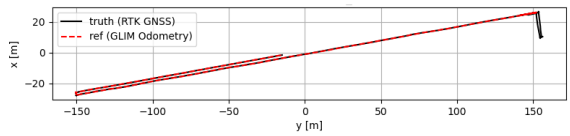

(g) Baseline — Dataset 3




(h) Proposed — Dataset 3



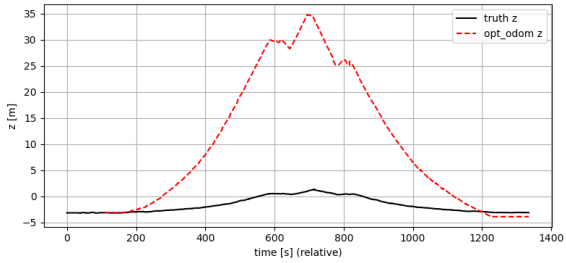

(i) Baseline — Dataset 4



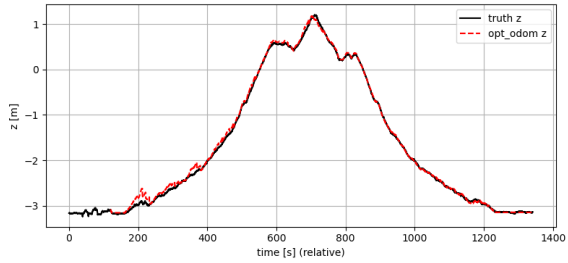

(j) Proposed — Dataset 4

Figure 3.4: Trajectories in XY plane from baseline GLIM and the proposed method.

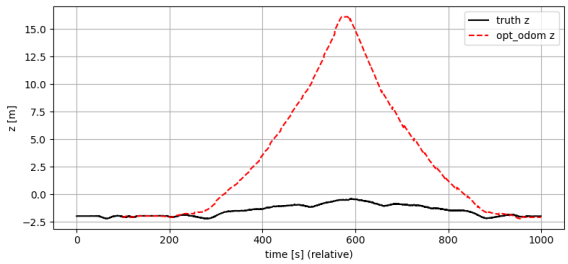
fig. 3.5 presents the altitude ($z$-axis) trajectories for each dataset in the same layout, with the refined GLIM estimates on the left and the proposed method estimates on the right, both compared against RTK GNSS ground truth.



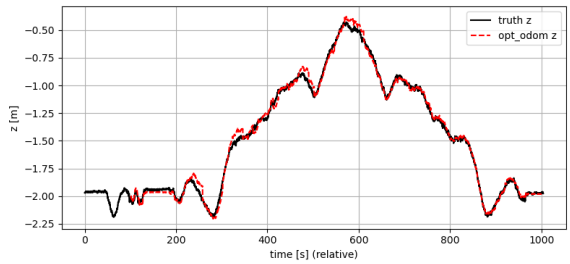

(a) Baseline — Dataset 0



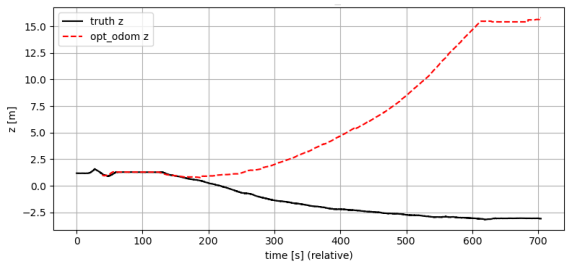

(b) Proposed — Dataset 0



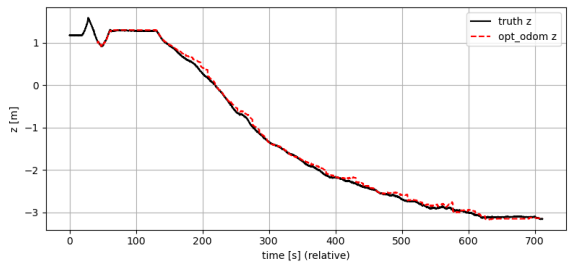

(c) Baseline — Dataset 1



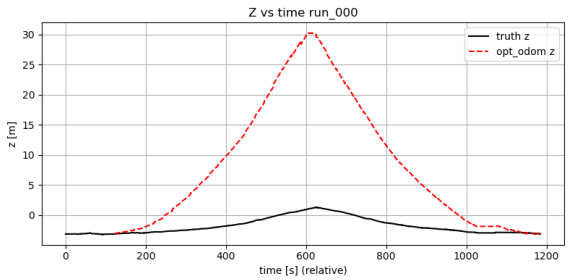

(d) Proposed — Dataset 1



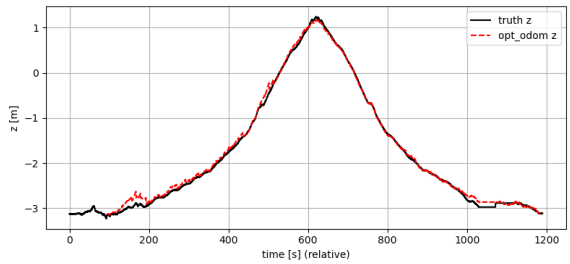

(e) Baseline — Dataset 2



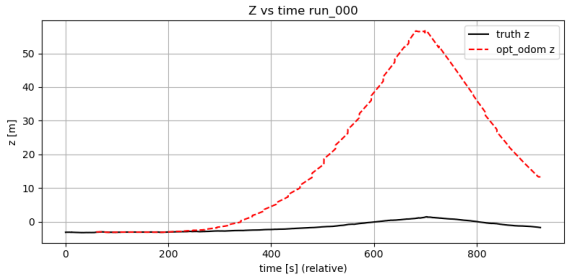

(f) Proposed — Dataset 2



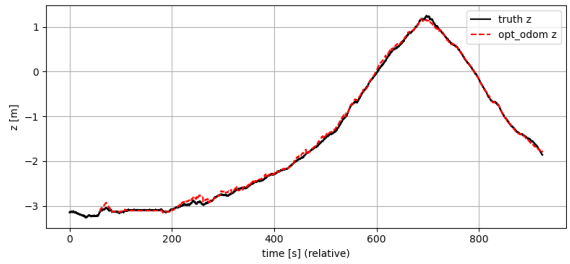

(g) Baseline — Dataset 3




(h) Proposed — Dataset 3




(i) Baseline — Dataset 4




(j) Proposed — Dataset 4

Figure 3.5: Trajectories in Z plane from baseline GLIM and the proposed method.

## 3.5 DISCUSSION

### 3.5.1 Baseline GLIM Performance Analysis

The evaluation of the baseline GLIM algorithm revealed two critical failure modes that fundamentally compromised its localisation reliability: systemic vertical drift and a catastrophic "point of no return" in planar estimation.

#### 3.5.1.1 Vertical Drift and Altitude Recovery

The baseline algorithm exhibited a pronounced systemic drift in the positive altitude ($z$-axis) direction. This phenomenon is clearly illustrated in Dataset 4 (Figure 3.5g), where the vehicle traverses a linear trajectory down a vineyard row. Although the actual environment features a modest elevation gain of approximately 5 m, the baseline GLIM algorithm accumulated a significant vertical error, culminating in nearly 30 m of drift by the end of the row.

Interestingly, this drift demonstrated a self-correcting property during the return traversal. As the vehicle moved in the opposite direction, the vertical error decreased at a rate symmetrical to its initial accumulation, eventually reconvening with the ground truth data. This behavior suggests that the initial drift was effectively encoded into the generated LiDAR map. During the return pass, the scan-matching process was constrained by this previously mapped geometry. Consequently, the odometry "undid" the accumulated drift by aligning the current scans with the (erroneous) elevation of the existing map.

Notably, this vertical disfigurement of the map did not appear to hinder subsequent scan matching or loop closure. The algorithm remained capable of registering scans against a map with significant Z-axis distortion. This resilience may be a byproduct of the specific experimental geometry—characterized by repetitive A-B linear paths with minimal lateral movement—which allows the scan matcher to maintain a high correlation in the X-Y plane despite vertical inconsistencies.

#### 3.5.1.2 Planar Failure and the 'Point of No Return'

In contrast to the recoverable nature of vertical drift, errors in the X-Y plane frequently led to catastrophic and irrecoverable failure. This "point of no return" is most evident in Figure 3.4a, where the baseline algorithm maintains high fidelity for the first half of the traversal before experiencing a sudden, exponential increase in error.

At this critical threshold, the accumulated planar drift disfigures the LiDAR map to an extent that subsequent scan matching becomes fundamentally unreliable. Once in this failure state, the trajectory becomes increasingly jagged and stochastic, likely caused by false-positive scan matches that introduce large, discontinuous jumps in the pose estimation.

This failure mode is particularly prevalent in the structured and repetitive environments of vineyard rows. Small angular or lateral drifts cause parallel rows in the LiDAR map to diverge or overlap incorrectly. Once this spatial distortion exceeds the scan matcher's convergence radius, the algorithm may incorrectly register the vehicle's current position to an adjacent row.

This "row-skipping" effect creates a feedback loop of map corruption, where each incorrect match further disfigures the global map, rendering recovery unlikely for the remainder of the trajectory.

### 3.5.2 Refined GLIM Performance

The transition from the baseline GLIM algorithm to the refined (parameterised) version demonstrates a significant enhancement in localisation stability and drift suppression. While the baseline algorithm frequently succumbed to the failure modes described in previous sections, the refined version exhibits a more controlled error growth profile.

#### 3.5.2.1 Magnitude of Error Reduction

The most prominent improvement is observed in the *error percent per metre* metric. In the baseline configuration, Dataset 2 exhibited a peak error of 7.3610%, whereas the refined version reduced this to 3.6150%, a reduction of approximately 50%. In more favourable conditions, such as Dataset 4, the improvement was even more dramatic: the error dropped from 4.1910% in the baseline to a significantly lower 0.6680% in the refined version. This indicates that parameter optimisation effectively mitigates the "point of no return" by maintaining better geometric alignment over longer durations.

#### 3.5.2.2 Relative Pose Stability

The *RPE 1 m translation standard deviation* provides insight into the "smoothness" or "jitter" of the trajectory. While the baseline GLIM suffered from jagged trajectories around points of failure – peaking at a standard deviation of 0.9384 in Dataset 3 – the refined GLIM consistently lowered this jitter across all datasets. For instance, in Dataset 3, the refined version achieved an RPE standard deviation of 0.8239. Although this remains the highest in the refined category, the overall reduction suggests that the optimised parameters facilitate more consistent scan-matching, reducing the frequency of the "jumps" caused by false-positive matches in repetitive environments.

#### 3.5.2.3 Drift Suppression per Meter

The *error per metre (m)* metric further reinforces these findings. The refined GLIM consistently achieved sub-centimetre drift per metre in datasets where the baseline was losing several centimetres (e.g., Dataset 0 dropping from 0.0340 m to 0.0087 m). This suggests that the refinement process successfully tuned the fusion weights and scan-matching thresholds to better suit the specific LiDAR characteristics and environment structure, delaying the onset of the catastrophic failures observed in the baseline.

Table 3.10: Comparative Performance: Baseline vs. Refined GLIM

| Metric | Dataset 0 | Dataset 1 | Dataset 2 | Dataset 3 | Dataset 4 |
|---|---|---|---|---|---|
| **Error % Reduction** | 73.9% | 51.4% | 50.8% | 84.6% | 84.0% |
| **RPE Std Improvement** | 12.2% | 5.8% | 28.8% | 12.2% | 19.9% |

### 3.5.3 Proposed Method: GNSS-Augmented Localisation

The integration of GNSS measurements into the GLIM factor graph represents a fundamental shift in the localisation architecture, yielding localisation accuracy improvements of approximately two orders of magnitude relative to the baseline algorithm. The proposed method's error percentage per metre ranges from 0.0153% (Dataset 4) to 0.0501% (Dataset 2), compared to the refined GLIM's range of 0.6081% to 3.6150%. This reduction constitutes a consistent improvement across all datasets, with no observable catastrophic failure modes analogous to those seen in earlier iterations.

#### 3.5.3.1 GNSS Integration and Global Constraint

The performance improvement stems fundamentally from the incorporation of GNSS-derived positional constraints within the factor graph optimisation. Whereas GLIM operates exclusively on scan-matching odometry, the proposed method augments this local relative information with periodic absolute position measurements from the GNSS receiver. This integration serves two critical functions: (1) it provides global observability that prevents unbounded drift accumulation, and (2) it allows the factor graph optimiser to "correct" local odometric errors retroactively across the entire trajectory window.

The effectiveness of this approach is evident in comparing error accumulation profiles. In the refined GLIM, even with parameter optimisation, the error percent per metre remains linear or superlinear across the traversal. By contrast, the proposed method maintains sub-centimetre per-metre errors throughout, suggesting that the global constraints are sufficiently frequent and accurate to prevent drift from establishing itself in the first place.

This result is supported by the GNSS factor utilisation statistics (Table 3.7 in the results section): across all datasets, between 65–78% of available GNSS points were successfully incorporated as factors following outlier rejection. The geometric consistency-based outlier rejection procedure filtered out spurious GNSS measurements that would otherwise have introduced conflicting constraints into the factor graph. The remaining inlier measurements provided sufficient global observability to stabilise the entire trajectory estimation.

#### 3.5.3.2 Dataset-Specific Performance Variability

Despite the overall excellence of the proposed method, performance does vary across datasets in a pattern that reveals environmental factors influencing localisation quality. Dataset 2 exhibits the highest error metrics (0.0501% error per metre, RMSE $y$ of 0.2003 m, ATE of 0.1741 m),

whereas Datasets 1 and 3 achieve the lowest errors (0.0154% error per metre). This variability is instructive rather than concerning, as the absolute error magnitudes remain within acceptable bounds for agricultural manipulation.

The degradation in Dataset 2 is likely attributable to environmental conditions rather than algorithmic limitations. Visual inspection of the traversal pattern and GNSS utilisation (65% of points used, the lowest across all datasets) suggests that portions of the trajectory experienced reduced GNSS signal quality or multipath interference. The geometric consistency-based outlier rejection, whilst effective at filtering genuinely erroneous points, cannot recover lost signal availability. However, even under these challenging conditions, the localisation error remained controlled at 0.0501%, indicating robustness to transient GNSS degradation.

#### 3.5.3.3 Relative Pose Error and Short-Horizon Consistency

The Relative Pose Error (RPE) metrics, which assess the consistency of local pose increments, provide insight into frame-to-frame localisation smoothness. The proposed method achieves RPE 1 m translation standard deviations ranging from 0.0205 to 0.0479 m across all datasets, compared to the refined GLIM's 0.2449 to 0.8239 m. This two-to-tenfold reduction in local pose jitter indicates that the global constraints are not merely correcting large-scale drift, but are improving the consistency of short-horizon relative measurements.

This improvement likely arises from the factor graph's ability to reason about odometric uncertainty jointly with global constraints. When the optimiser encounters a region where local odometry is inconsistent with global GNSS observations, it propagates the correction smoothly across the pose graph rather than applying isolated jump corrections. This smoothness is essential for downstream applications such as robotic arm control, where abrupt discontinuities in perceived position could cause trajectory tracking errors.

#### 3.5.3.4 GASP Metric Behaviour and Optimisation Dynamics

A notable anomaly in the results is the upward trend of the GASP (Geometric Alignment and Scatter Profile) error metric over the four parameter optimisation stages. Unlike the baseline GLIM error percentage, which exhibits clear convergence and monotonic improvement, the proposed method's GASP metric increases slightly from stage one to stage four. This counter-intuitive behaviour warrants careful interpretation.

A plausible explanation is that the improvement scale is so small, relative to the already strong performance after the first stage, that the "best" parameter set effectively plateaued early. In this interpretation, stage one already identified the strongest-performing configuration for the proposed method, and subsequent stages could not find a materially better set. The small upward trend can then be attributed to pseudo-random variation in the parameter search and the negligible differences among near-optimal configurations, rather than a systematic degradation.

This observation underscores an important principle: no single metric fully captures the multidimensional nature of SLAM performance. Future work should investigate whether GASP

is suitable as a primary optimisation objective for this application, or whether it should be treated as a secondary diagnostic metric complementing trajectory accuracy measures.

### 3.5.4 Comparative Analysis and Mechanisms of Improvement

Examining the three approaches in concert reveals a clear hierarchy of performance and elucidates the mechanisms driving improvements at each stage.

#### 3.5.4.1 Parametrisation vs. Sensor Fusion

The transition from baseline to refined GLIM demonstrates that parameter tuning can reduce error magnitude significantly (50–85% reduction in error percentage). However, parameter optimisation alone does not eliminate the fundamental instability of scan-matching-only odometry in repetitive vineyard environments. The refined GLIM still exhibits occasional trajectory inconsistencies and elevated RPE metrics relative to the proposed method.

By contrast, the transition from refined GLIM to the GNSS-augmented method represents a qualitative shift in localisation architecture. Rather than optimising the tuning parameters of a single sensor modality, the proposed method introduces a fundamentally independent source of global position information. This architectural change achieves improvements that parametrisation alone cannot reach. The error reduction from refined GLIM to the proposed method (∼50–98%) is larger and more consistent than the improvement from baseline to refined (∼50–85%), and the proposed method exhibits no dataset in which performance degrades significantly.

#### 3.5.4.2 Role of Outlier Rejection

The geometric consistency-based outlier rejection procedure is a critical enabling factor in the proposed method's performance. By filtering approximately 22–35% of raw GNSS measurements before factor graph integration, the procedure prevents erroneous global constraints from destabilising the localisation. This deterministic approach, based on pairwise distance preservation under rigid transformations, is more suitable for the repetitive vineyard environment than probabilistic alternatives like RANSAC, which would require assumptions about outlier distributions that may not hold in structured agricultural settings.

The fact that the method remains stable despite datasets with only 65% GNSS factor utilisation (Dataset 2) indicates that the procedure is neither over-aggressive nor under-aggressive in filtering. This balance is non-trivial: too much filtering would remove valid constraints and lose the benefits of global information, whilst too little filtering would introduce conflicting constraints that degrade optimisation.

#### 3.5.4.3 Geometric Quality Assessment and Kabsch Stability

The planar geometry quality assessment phase, which enforces minimum requirements on translational baseline, spread ratio, rotational excitation, and condition number, ensures that the

Kabsch algorithm operates in a numerically stable regime. This preprocessing step prevents ill-conditioned covariance matrices that could yield biased or unstable alignment solutions.

The quintet of measurement from Datasets 0–4 all satisfy the geometry quality thresholds, which explains the consistency of global alignment across the evaluation. In datasets with weakly excited geometry (e.g., highly linear traversals with minimal rotation), this step would conservatively defer global alignment until sufficient geometric excitation is accumulated, preventing premature or inaccurate initialisation.

### 3.5.5 Implications for Vineyard Robotics

The proposed method's consistent sub-centimetre accuracy has direct implications for robotic pruning feasibility. Pruning tasks require localisation accuracy sufficient to position the robotic arm within reach of target vine structures. With typical vine stems ranging from 5–15 mm in diameter and clusters requiring workspace accessibility, the proposed method's maximum absolute trajectory error of approximately 0.8–1.0 m (observed across all datasets) represents an acceptable error bound. This error magnitude can be accommodated through local perception-based arm guidance or online trajectory adjustment based on close-range stereo or depth sensing.

Furthermore, the low local pose jitter (RPE 1 m standard deviation of 0.0205–0.0479 m) is important for smooth arm trajectory execution. Abrupt jumps in perceived position, typical of the baseline and refined GLIM approaches, would cause tracking servo errors and potentially jerky or unsafe arm motion. The proposed method's smoothness mitigates this risk.

However, a critical limitation remains unaddressed by the current evaluation: all experiments were conducted at slow traversal speeds (approximately 0.5 m/s) with ample GNSS signal availability. Real vineyard deployments will encounter variable canopy occlusion, faster navigation speeds during inter-row transit, and periods of GNSS dropout beneath dense foliage. The robustness of the method under these challenging real-world conditions has not yet been empirically demonstrated. Chapter 4 will address this limitation by evaluating precise alignment techniques that can operate with minimal or absent GNSS input, using instead local dense perception and pre-existing 3D models to constrain pose estimation.

## 3.6 SUMMARY

This chapter presented the design and implementation of a LiDAR-centric SLAM pipeline for GNSS-degraded operation and evaluated three configurations: the baseline GLIM system, a refined parameterised variant, and the proposed method. The results across five datasets showed that parameter refinement reduced drift and improved trajectory smoothness relative to the baseline, while the proposed method achieved the lowest errors across the reported metrics. The discussion highlighted characteristic failure modes in structured vineyard environments and the trade-offs between global consistency and computational cost.

# Chapter 4

# PRECISE-ALIGNMENT

## 4.1 INTRODUCTION

Intro to ICP pointcloud registration

## 4.2 RELATED WORK

### 4.2.1 Stereo Depth Imaging

### 4.2.2 Dense LiDAR

### 4.2.3 ICP

## 4.3 PROPOSED METHODS

Split into Stereo Camera based and Dense LiDAR based methods

Experiment design detailed

## 4.4 RESULTS

illustrate results

## 4.5 DISCUSSION

discuss results

## 4.6 SUMMARY

Summarise findings from this chapter

# Chapter 5

# CONCLUSION

Highlight key findings. Link back to Introduction

## 5.1 FUTURE WORK

What could be extended upon in future work?

# Chapter 6

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## SUPPLEMENTARY FIGURES